


Optimization under Uncertainty and Applications in Energy and Power Systems

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NC STATE UNIVERSITY | OPERATIONS RESEARCH GRADUATE PROGRAM

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Talk delivered at Politecnico di Torino

Politecnico di Torino

Overview

- Introduction
- Stochastic Programming Basics
- Multi-stage Stochastic Linear Programs (MSLPs)
- Applications of Stochastic Programming in Energy & Power
- Sampling-based Decomposition (SBDA)
- Final Comments

Introduction

About Myself

- Born in Mogi Mirim, SP - Brazil
- Academic Background
  - B.S. in EE (2005)
  - M.Sc. in EE (2007)
  - Ph.D. in ORIE (2011)
- Appointments
  - Consultant & Partner (2006 - 2013)
  - Asst. Professor (2013 - 2015)
  - Res. Asst. Professor (2015 - 2017)
  - Assistant Professor (2017 - 2023)
  - Associate Professor (2023)

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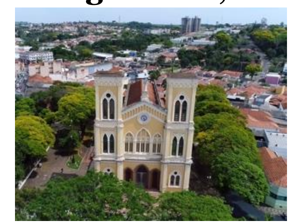
NORTH CAROLINA STATE UNIVERSITY NC STATE UNIVERSITY Department of Civil, Construction, & Environmental ENGINEERING Operations Research Program

Mogi Mirim, SP



Itajubá, MG

Austin, TX

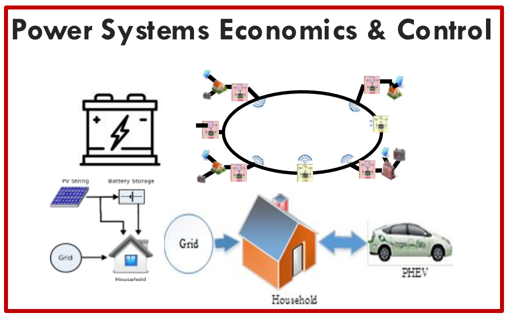
Raleigh, NC

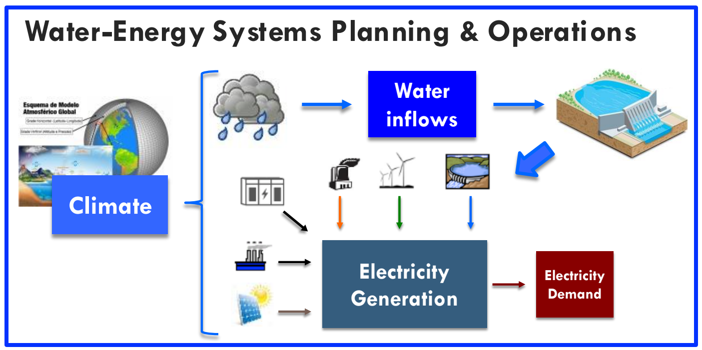
About my Research

One of my goals is to better use theoretical developments and apply them to model-based analysis that informs and improves real world decision-making

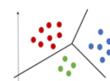
Decision-making under uncertainty

Computational Optimization



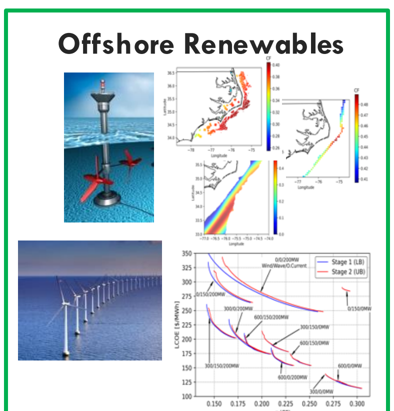


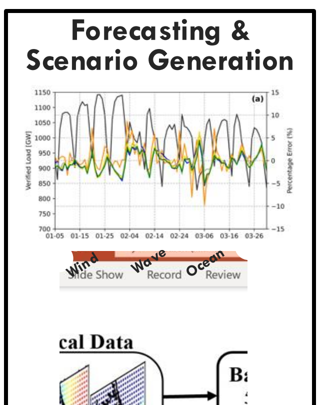
Model Design

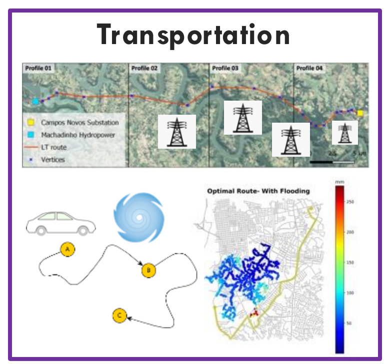


Clustering Analysis

Leverage advanced ***computational optimization*** *and* ***predictive analytics methods*** *to* ***improve decision-making in complex systems***







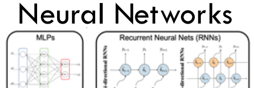
Over the years I have been applying my expertise to ***civil engineering systems***

Data Envelopment Analysis (DEA)

Multidisciplinary problems*, I look to collaborate with other researchers*

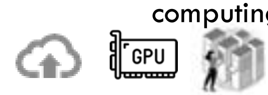
Portfolio Optimization

Neural Networks

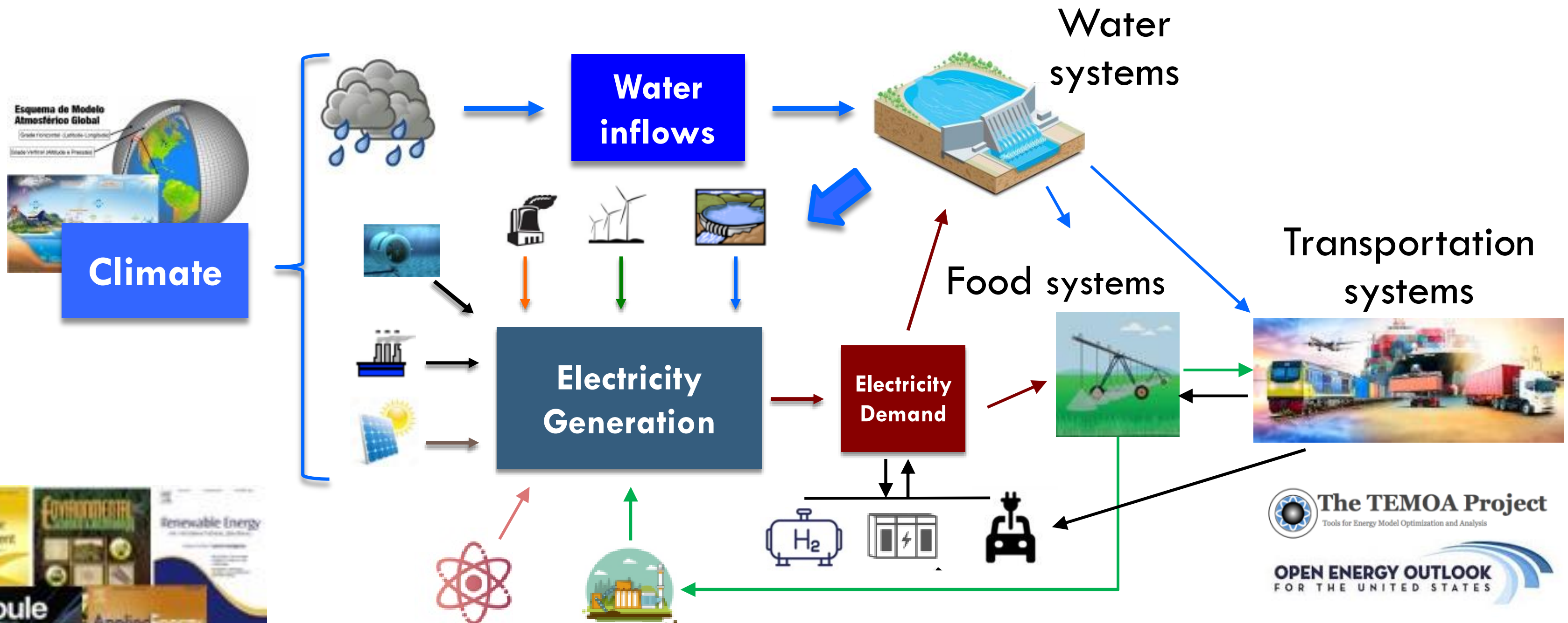


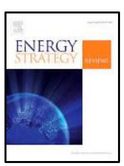
Predictive Analytics

High performance computing

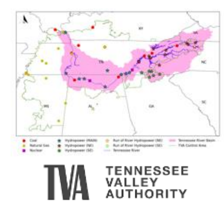


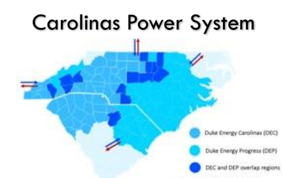
Water, Energy, and Power Systems Planning & Operations Under Future Uncertainty

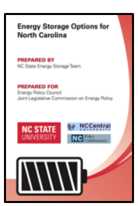




Patankar, N., Eshraghi, H., de Queiroz, A. R., & DeCarolis, J. F. (2022). Using robust optimization to inform US deep decarbonization planning. Energy Strategy Reviews, 42, 100892



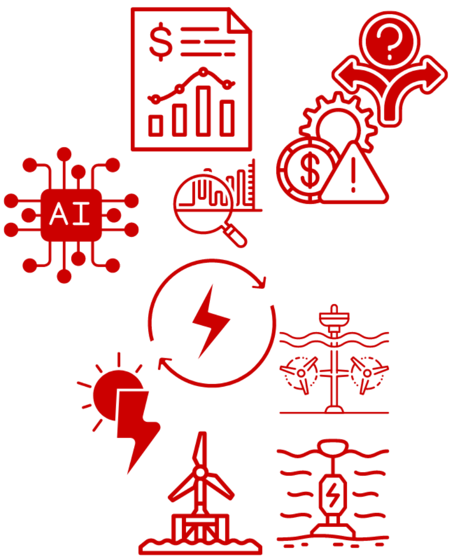




Hafiz, F., Awal, M. A., de Queiroz, A. R., & Husain, I. (2020). Real-time stochastic optimization of energy storage management using deep learning-based forecasts for residential PV applications. IEEE Transactions on Industry Applications, 56(3), 2216-2226

The Energy and Power Sector

- The energy sector is crucial in the global economy
  - **Annual investments** exceed **US$2.8 trillion worldwide**
- Concerns about **climate change** and the **search for sustainability** have driven renewable sources
- **Renewable sources** have **led new capacity additions** in the world since 2012 (84%: 2021 and 95%: 2026)



- The **goal to decarbonize systems** (zero CO_2 by 2050)
  - According to IRENA, energy generation is expected to triple by 2050
  - Annual global investments in the sector could reach **US$5 trillion by 2030**

A Generation Paradigm Shift



- Historically
  - Centralized generation
  - Fossil fuel, nuclear, large hydropower
- Current Situation
  - More dynamic
  - More distributed
  - More renewable generation

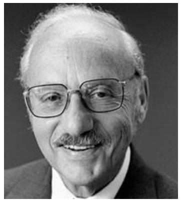

- But some aspects remain unchanged
  - Need for balancing supply and demand
  - Reliable and Robust

Modeling and decision-making under uncertainty are essential for operational and planning actions

Stochastic Programming Basics

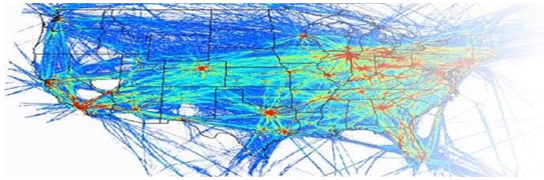
Optimization Under Uncertainty

- **Stochastic programing** (SP) is used as a tool for modeling **optimization** problems under **uncertainty**
- This operations research area was born in the 1950s "Linear programming under uncertainty" (**Dantzig, 1955**)



Power generation scheduling

Financial markets

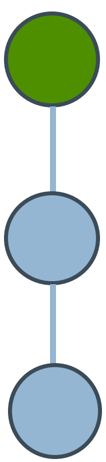


Transportation planning

- Real problems constantly include parameters that are unknown when decisions should be made
- SP models rely on the assumption that **probability distributions** are known or can be estimated
- There are different classes of SPs, and our focus is on **multi-stage stochastic linear programs** (SLP-t)
- Decisions have to be made at different time stages and there is a dynamic link between stages

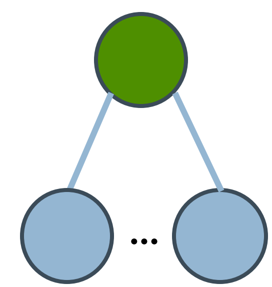
Introduction to Stochastic Optimization

- Deterministic models assume that all parameters are known with certainty
- Stochastic optimization involves **decision-making under uncertainty**, i.e., the problem data is not deterministic but described by probability distributions
- Key concepts



Stages of decision-making

Uncertainty modeling



Key concepts

Stages of decision-making

- **Single-stage models:** Decisions are made once under uncertainty
- **Two-stage models**: First-stage decisions are made before uncertainty is realized, and second-stage (recourse) decisions are made after
- **Multistage models**: A series of decisions are made over time as new information is revealed

Uncertainty modeling

- **Scenarios:** Representing **uncertainty through discrete scenarios** (e.g., different weather patterns)
- **Probabilities:** Assigning **likelihoods to scenarios** or using probability distributions (e.g., normal distribution for demand)

Recourse in Stochastic Optimization

- Recourse is the ability to adjust decisions once uncertainty has been revealed
- It's a key feature of two-stage and multistage stochastic optimization models
- **First-stage vs. second-stage decisions**: First-stage decisions are made with limited information, while second-stage decisions are "recourse" actions taken to correct or adjust after the uncertainties unfold

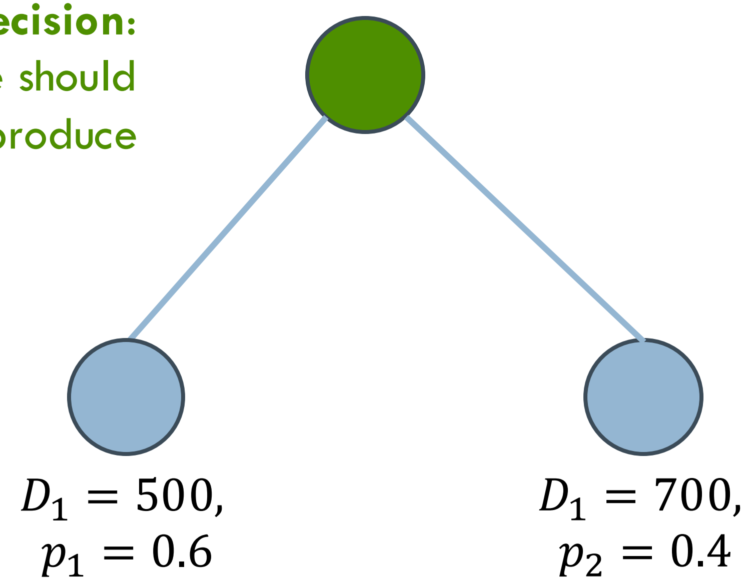
Example 1: Problem Description

- Suppose we have to decide on the number of water heaters we manufacture
  - The production cost of each unit is $2
  - Customers demand is random and represented by a discrete distribution
    - 500 units with probability of 0.6 or
    - 700 with probability of 0.4
  - Customer demand must be met
  - We have the opportunity to buy the water heater from an external supplier. The purchase cost is $3
  - How much should we produce at the moment while we don't know the future demand?

Example 1: Problem visualization

First Stage decision:
x_1: number of units we should produce

Second Stage decision:
$y_{2,s}$: number of units we should purchase to meet demand



- Optimal solution is not the expected amount of demand!
- Optimal solution can be obtained through stochastic optimization

Example 1: Problem Formulation

$$\min_{x,y} \quad \mathrm{z} = 2\,\mathrm{x}_1 + \sum_{\mathrm{s}=1}^{2} [p_s \cdot 3 \cdot \mathrm{y}_{2,\mathrm{s}}]$$

s.t.

$$x_1 + y_{2,s} \geq D_s \qquad s = 1,2$$

$$x_1 \geq 0$$

$$y_{2,s} \geq 0 \qquad s = 1,2$$

Example 1: Problem Formulation

$$\min_{x,y} \qquad z = 2\,x_1 + \sum_{s=1}^{2} [p_s \cdot 3 \cdot y_{2,s}]$$

s.t.

$$x_1 + y_{2,s} \geq D_s \qquad s = 1,2$$

$$x_1 \geq 0$$

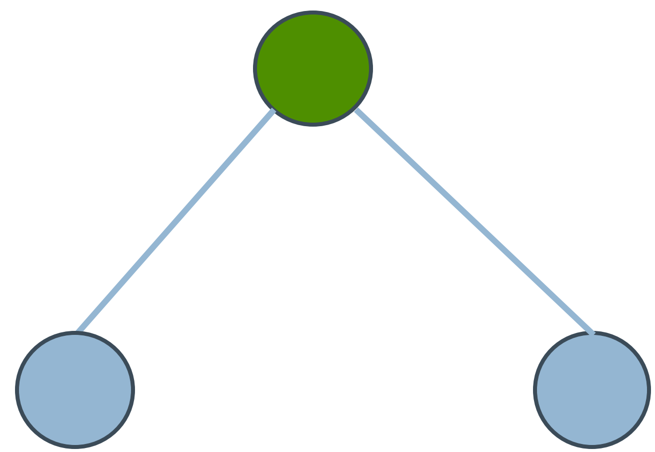
$$y_{2,s} \geq 0 \qquad s = 1,2$$

Optimal solution

$$x_1 = \mathbf{500}$$

$$y_{2,1} = \mathbf{0},\, y_{2,2} = \mathbf{200}$$

$$z = \mathbf{1240}$$



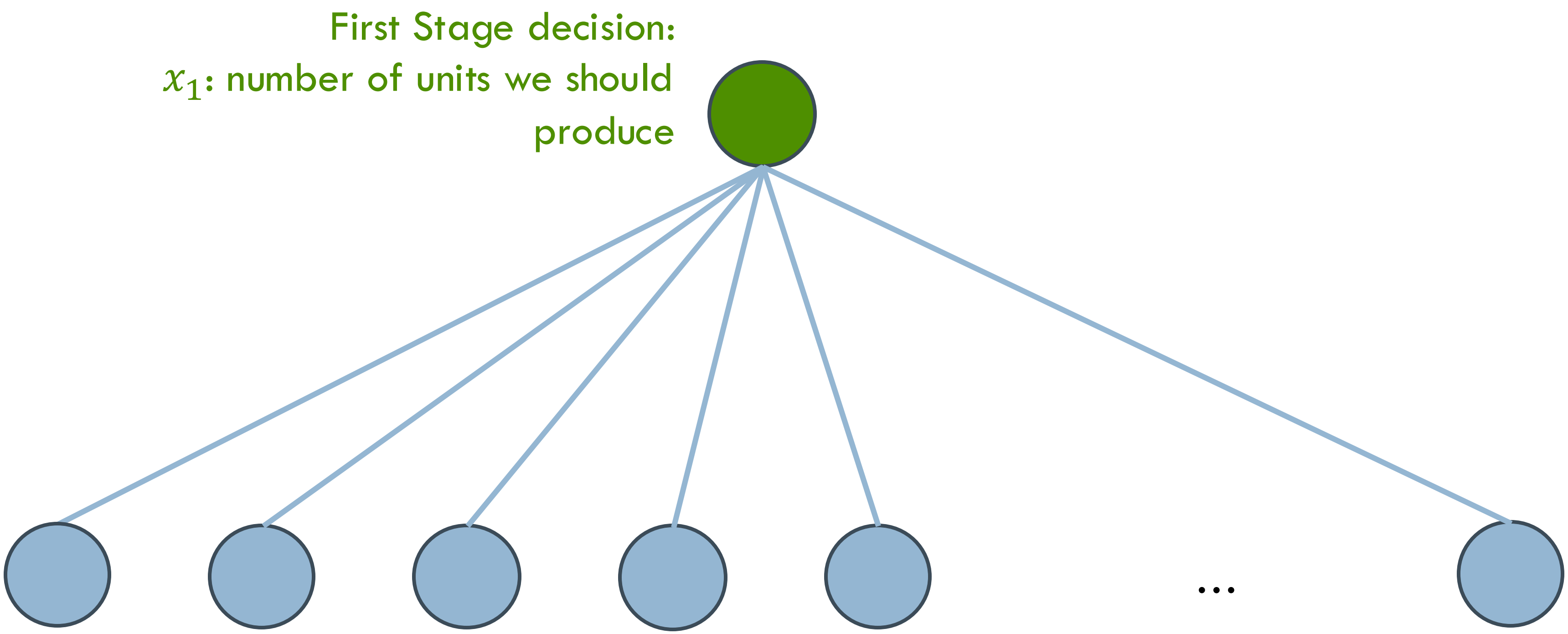
$D_1 = 500$
$z = 1000$

$D_1 = 700$
$z = 1600$

Example 2: Continuous Distribution

- Now imagine that instead of two discrete demand values (500 and 700 units), the demand is modeled as a continuous normal distribution with a mean of 600 units and a standard deviation of 100
- How would you represent the uncertainty in this case?

Example 2: Scenario Tree



- How many scenarios do we need?
- Would changing the scenarios change my objective function?

Multi-stage Stochastic Linear Programs (MSLPs)

Multi-stage Stochastic Programs with Recourse

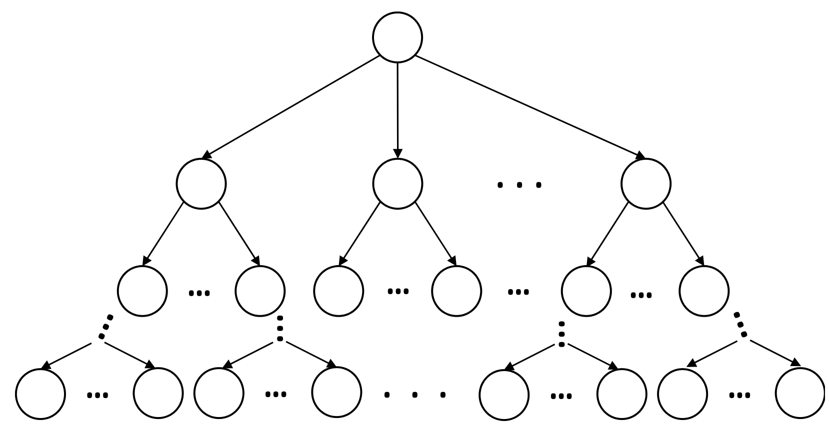
$$\min_{x_1} \; c_1x_1 + E_{\xi_2}\{\min_{x_2} c_2x_2 + E_{\xi_3|\xi_2}\left[\min_{x_3} c_3\, x_3 + \cdots + E_{\xi_T|\xi_{T-1}\ldots\xi_1}\left(\min_{x_T} c_Tx_T\right)\right]\}$$

$$\begin{aligned}
\text{s.t.}\quad & A_1x_1 - B_1x_0 && = b_1 \\
& A_2x_2 \quad - B_2x_1 && = b_2 \\
& \quad\vdots \qquad\qquad \ddots && \;\;\vdots \\
& A_Tx_T \qquad\quad - B_Tx_{T-1} && = b_T
\end{aligned}$$

$$\underline{x}_t \le x_t \le \bar{x}_t \;\forall t \in T$$

$\xi_t = \{c_t, A_t, B_t, b_t\} \quad \forall t \in T$

Represents the uncertainty vector



Example of a Scenario tree

Complete recourse: first stage solution does not yield an infeasible second stage problem

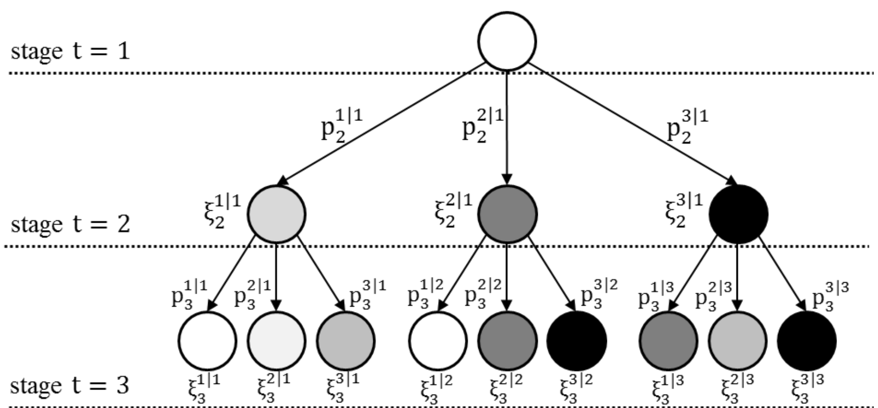
A General Multi-stage Stochastic Linear Program

We consider a general model such as:

$$\min_{x_1} \quad c_1 x_1 + E_{\xi_2|\xi_1} h_2(x_1, \tilde{\xi}_2)$$
$$\text{s.t.} \quad A_1 x_1 = B_1 x_0 + b_1$$
$$x_1 \geq 0$$

where, for $t = 2,\dots,T$

$$\min_{x_t} \quad c_t x_t + E_{\xi_{t+1}|\xi_1,\dots,\xi_t} h_{t+1}(x_{t-1}, \tilde{\xi}_{t+1})$$
$$\text{s.t.} \quad A_t x_t = B_t x_{t-1} + b_t$$
$$x_t \geq 0$$



Three-stage Scenario tree

x_t: stage t decision variables

A_t: constraint matrix

b_t: resource vector

c_t: cost vector

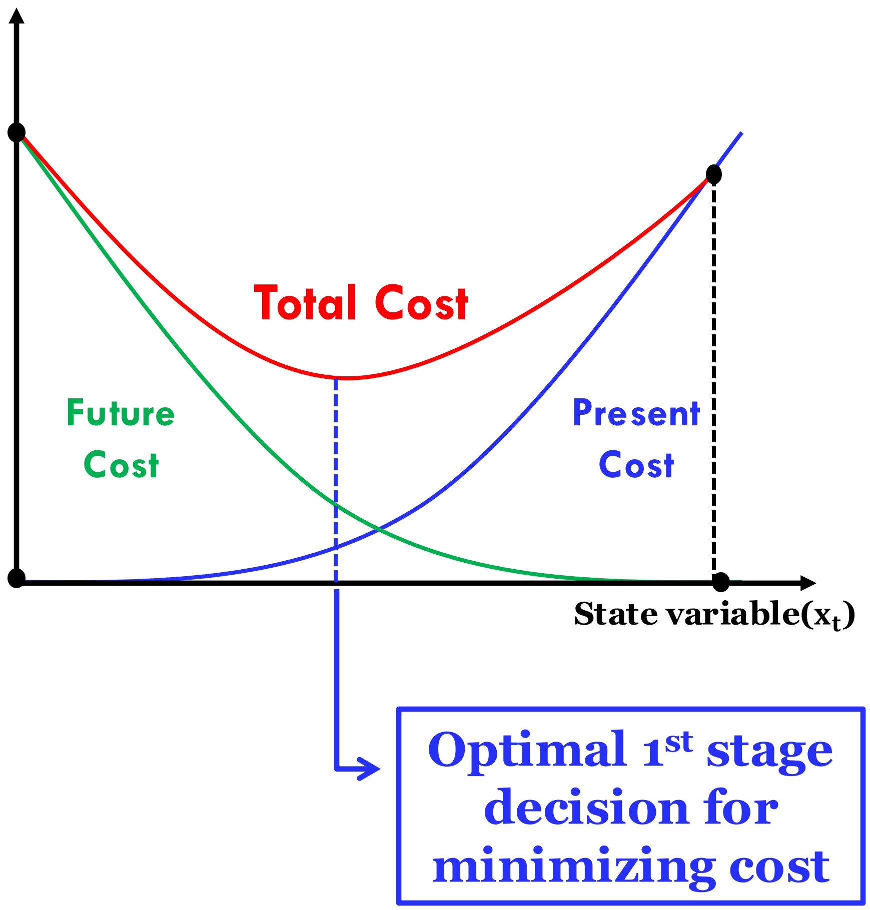
$B_t x_{t-1}$: transition from last stage

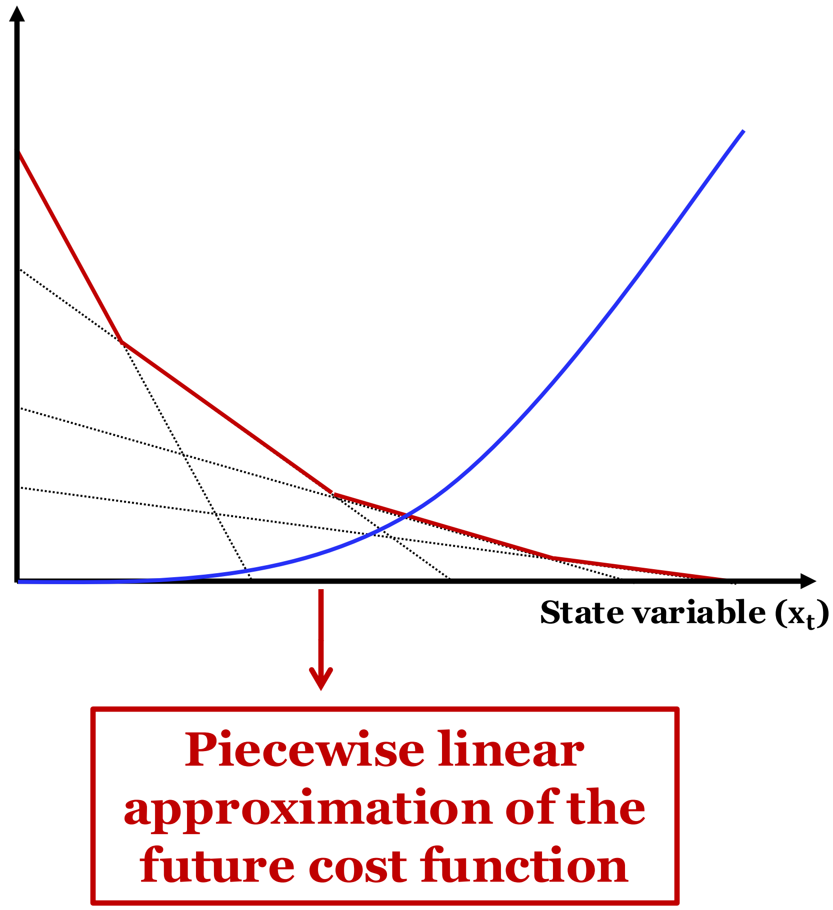
p: scenario probability

ξ_t: uncertainty vector at stage t

Model's Objective

- **Minimize total costs (present + future)**





Applications of Stochastic Programming in Energy & Power

Uncertainty and Metrics in MSLP

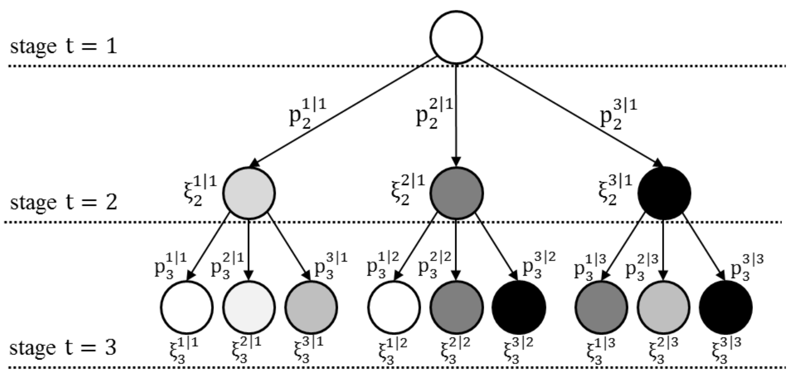
- **Uncertainty plays a fundamental role** in the definition of strong strategies which aim **to minimize the total combined expansion and operational costs** over a particular time horizon using mathematical models
- Understanding uncertainty is also a pre-requisite to correctly use such information in mathematical models **(Kann & Weyant, 00)**
- The representation of each uncertainty type will result in different models and different results and values that can be achieved with such models
- We aim to investigate long-term ESPP under uncertainty and what is the value of a MSLP representation of the problem instead of a deterministic
- Expected cost of perfect information **(EVPI)** **(Raiffa & Schlaifer, 61; Birge & Louveaux, 97)** → **how much to pay to eliminate uncertainty?**
- Value of the stochastic solution **(VSS)** **(Birge, 82)**

How much money the hedging strategy saves relative to the total cost obtained by an optimization model when uncertainty is ignored

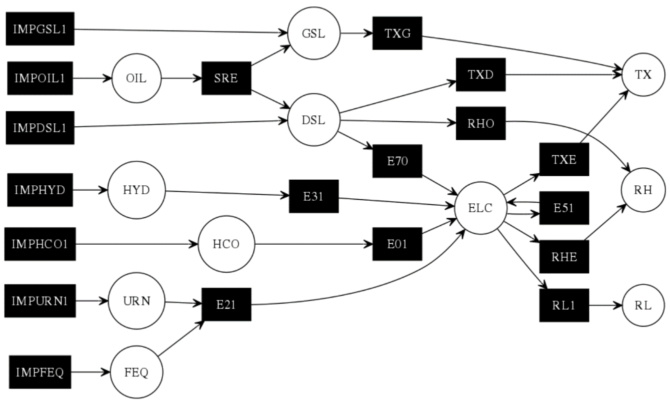
Uncertainty Representation

- Uncertainties may be represented by scenarios of:
  - Economic grow
  - Commodity/technology price trajectories **(Van der Weijde & Hobbs, 12)**
  - Demand realization **(Pineda and Morales, 2016)**
  - Technology reliability **(Hajipour et al., 2015)**
  - Policies related to greenhouse gases emissions **(Bistline & Weyant, 13; Park & Baldick, 15)**
  - Renewable generation penetration **(Munoz et al., 14)**
  - Renewable resources availability **(Gil et al., 15)**
  - Technological, economic, and policy-related **(Bistline, 15)**

ESPP as a MSLP



Ex. of a scenario tree



Ex. of ESSP network topology at a specific stage

$$z_{RP} = \min_{x_1} \; c_1 x_1 + \mathbb{E}_{\xi_2|\xi_1} h_2(x_1, \tilde{\xi}_2)$$

$$\text{s.t.} \;\; A_1 x_1 = B_1 x_0 + b_1$$

$$x_1 \geq 0$$

$\tilde{\xi}_t$ denote the random elements from (A_t, B_t, c_t, b_t) for $t = 2, \ldots, T$

where for $t = 2, \ldots, T$ the recourse function $h_t(x_{t-1}, \tilde{\xi}_t)$ can be viewed as:

$$h_t(x_{t-1}, \tilde{\xi}_t) = \min_{x_t} \;\; \tilde{c}_t x_t + \mathbb{E}_{\tilde{\xi}_{t+1}|\xi_t,\ldots,\xi_1} h_{t+1}(x_t, \tilde{\xi}_{t+1})$$

$$\text{s.t.} \;\; \tilde{A}_t x_t = \tilde{B}_t x_{t-1} + \tilde{b}_t$$

$$x_t \geq 0$$

The TEMOA Team Over Time, well part of…


Victor Faria


Matteo Nicoli


Cameron Wade


Jethro Ssengonzi


Qoriatul fitriyah


Daniele Larede


Mike Blackhurst


Binghui Li


Hadi Eshraghi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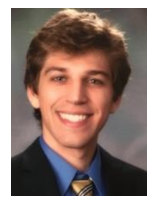

Dani Sodano


Dustin Soutendijk


Katie Jordan


Lucas Scianni


Jeff Thomas

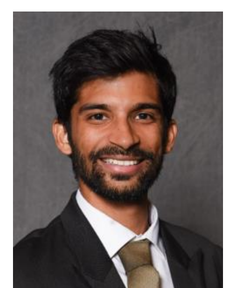

Joseph DeCarolis


Anderson de Queiroz


Jeremiah Johnson


Sankar Arumugam


Paulina Jaramillo

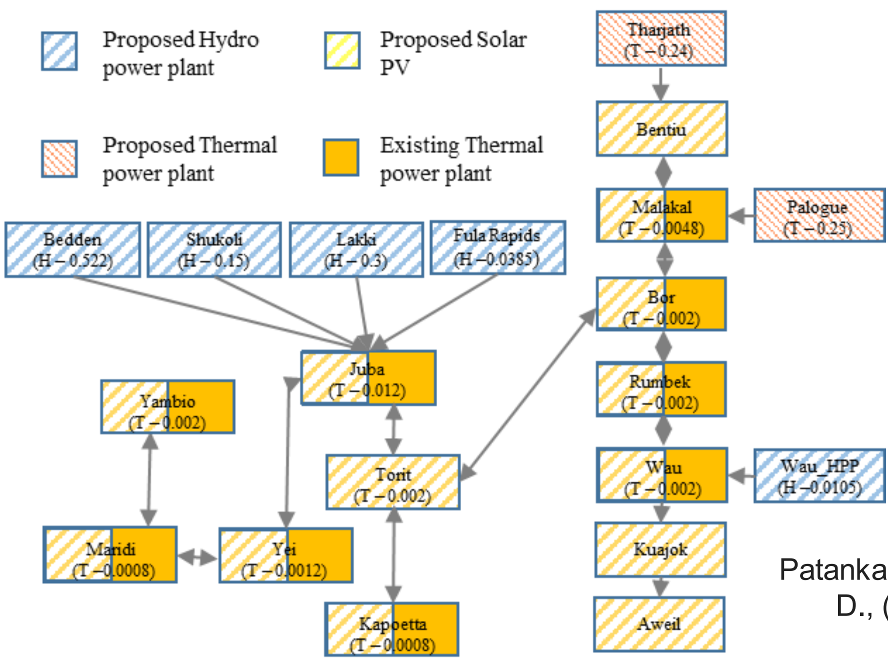

Aditya Sinha


Laura Savoldi

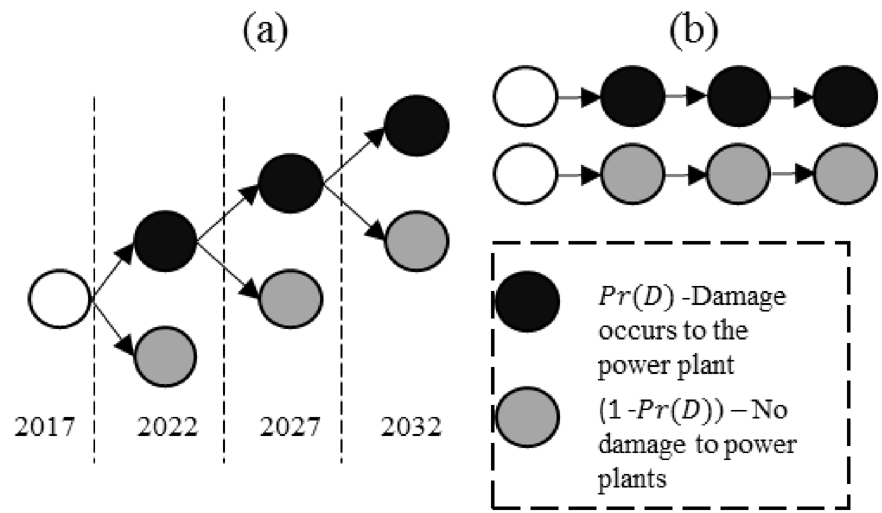

Daniel Posen

Building Conflict Uncertainty into Electricity Planning

- South Sudan has been subject to conflict since independence in 2011
- The total installed capacity was ~30 MW (serves 1% of the population)
- Past studies proposed building large scale hydro power plants (HPP)



Multi-stage Stochastic Optimization

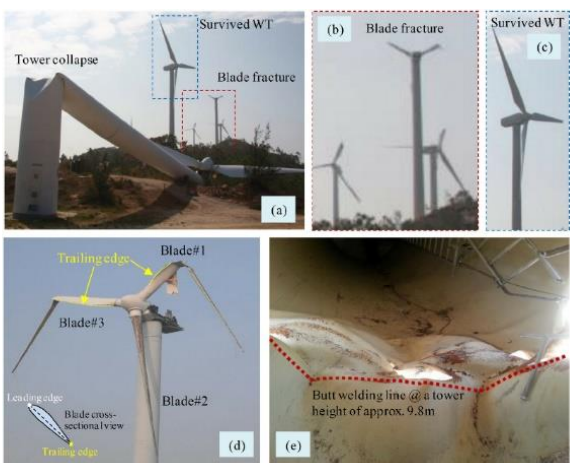


- Large plants such as hydro are vulnerable to damage during conflict
- A decentralized system might be less expensive considering conflict risk.

Patankar, N., de Queiroz, A.R., DeCarolis, J.F., Bazilian, M., Chattopadhyay, D., (2019) Building Conflict Uncertainty into Electricity Planning: A South Sudan Case Study, Energy for Sustainable Development, 49: 53-64

Risk of Hurricane Damage to the Power System

www.thecoastlandtimes.com



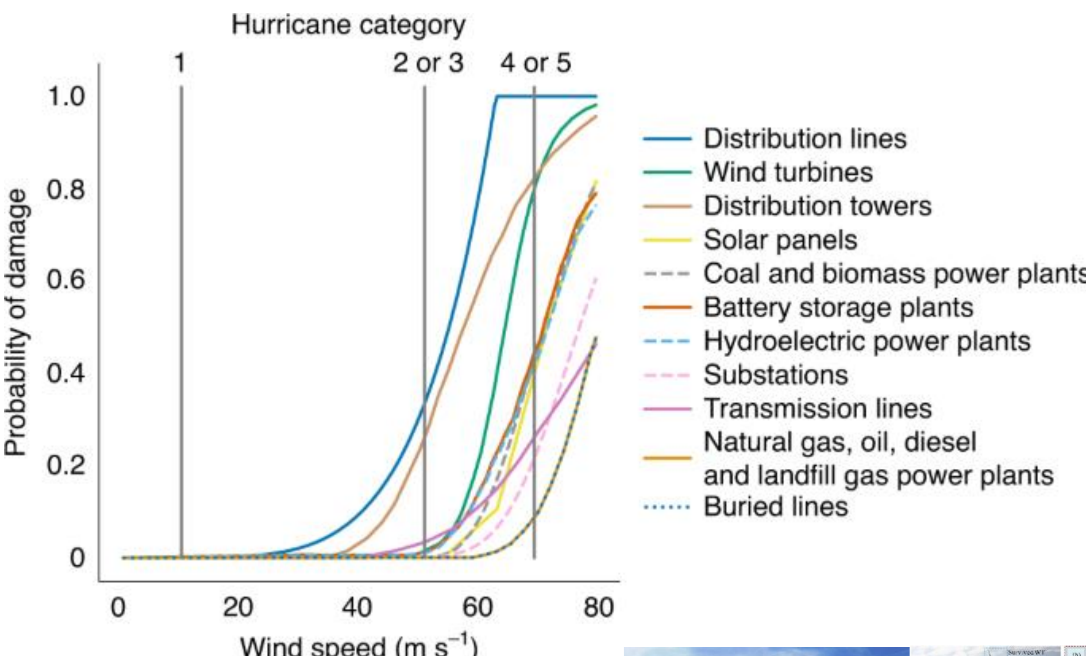
10.1016/j.engfailanal.2015.11.028

- The estimated cost to Duke Energy's NC customers for the storm repairs made during 2018-2019 due to Hurricanes Florence and Dorian account for more than US$1 billion

- Hurricane Irma (2017) is estimated to cost US$ 1.3 billion to Florida Power & Light customers and Hurricane Ian (2022) is estimated to cost US$1.1 billion

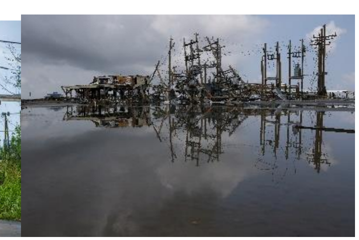
Risk of Hurricane Damage to the Power System

Fragility Curves (Bennett et al., 2021)



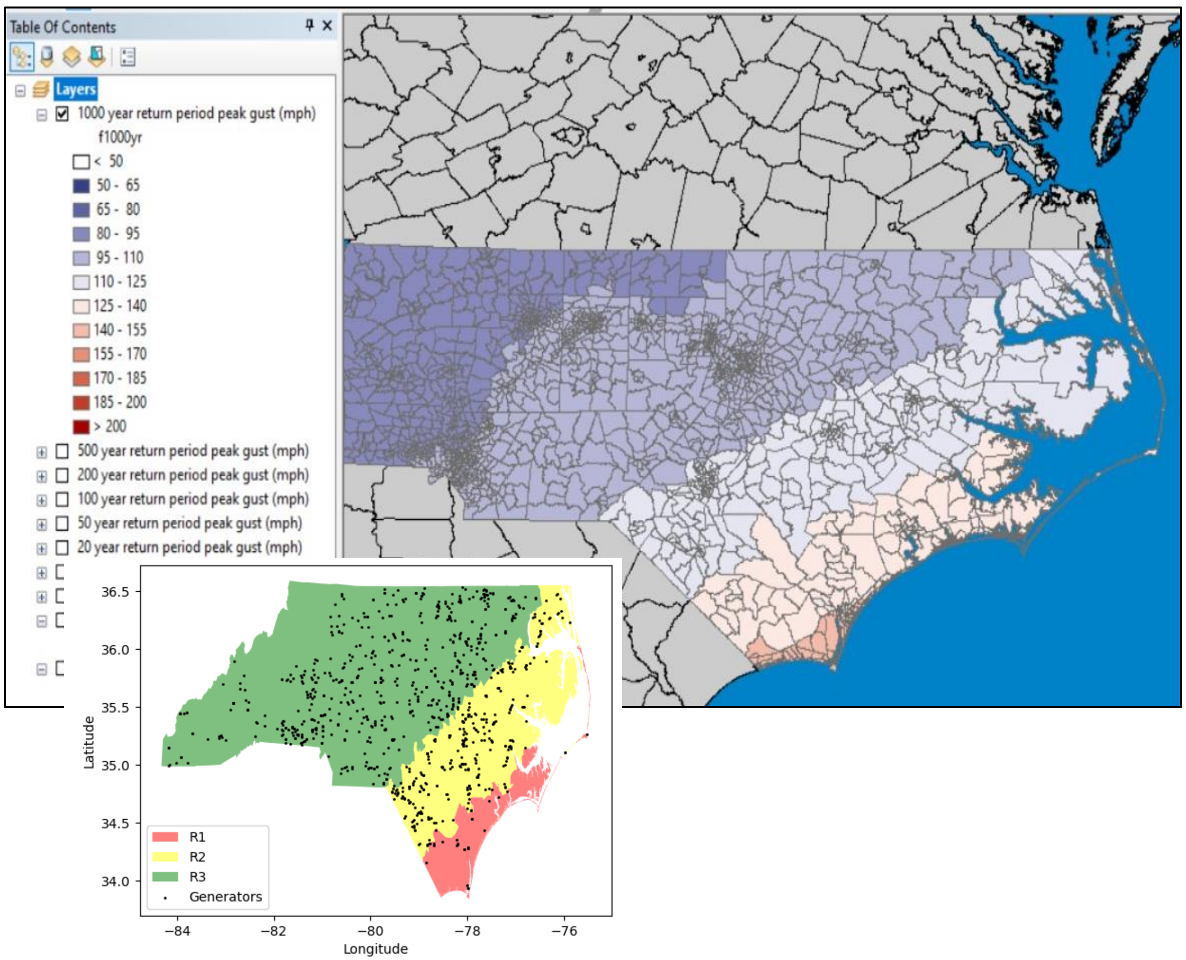
- Probability of damage to the power system infrastructure given different hurricane categories
- Storm damage can be represented as a fraction of existing capacity that is no longer usable to meet demand



Extreme Wind Conditions: Recurrence Periods

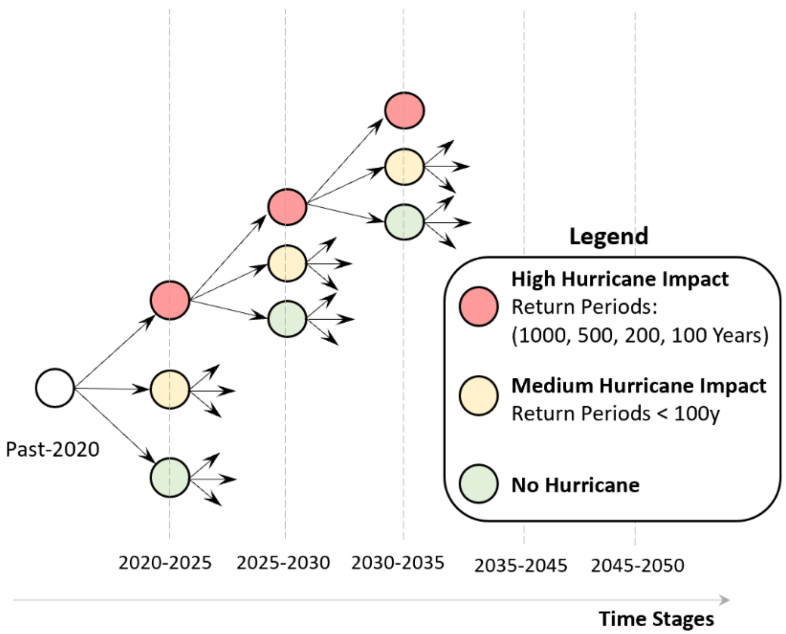
Hurricane Speed Statistics (FEMA - HAZUS)



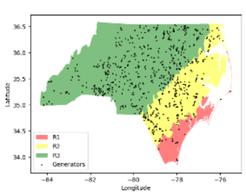
- HAZUS: Wind speed at different return periods
- For existing generation we can attribute specific extreme wind speed statistics if location is known
- For future generation we need to define a finite number of regions where the technology may be deployed

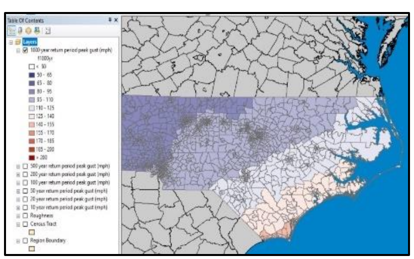
Scenario Tree Construction

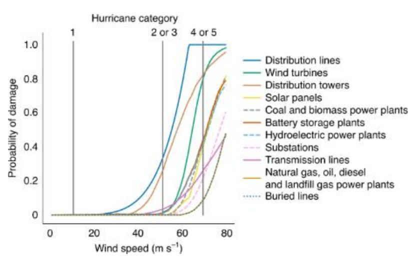
Scenario Tree



- Five time discretization; Five year intervals; Three scenarios: $\mathbf{3^5 = 243}$ **Possible Realizations**
- Three distinct scenarios: High, Medium and low hurricane damage, based on the recurrence periods from HAZUS
- The total technology damage at a given scenario is computed by taking the expectation of the damage caused by each of the hurricane return periods (weighed average)



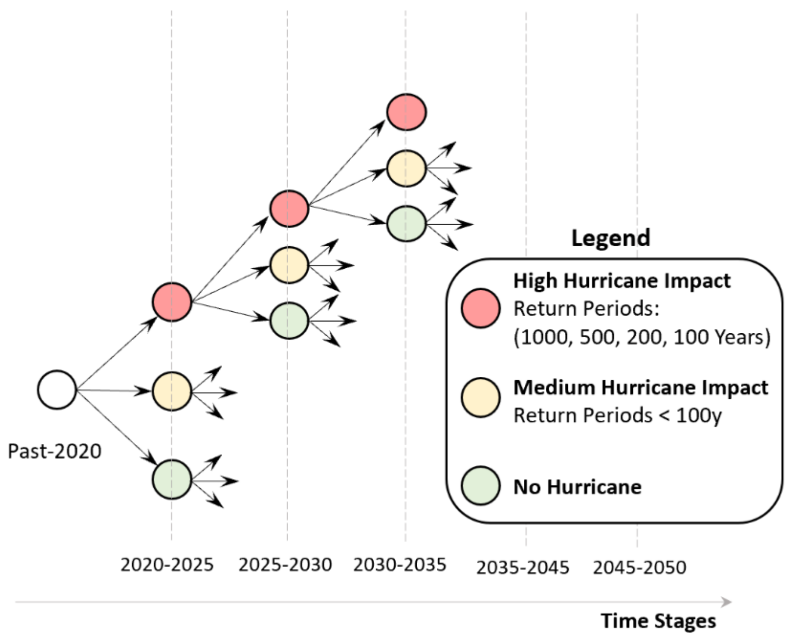




Damage at each scenario and time stage

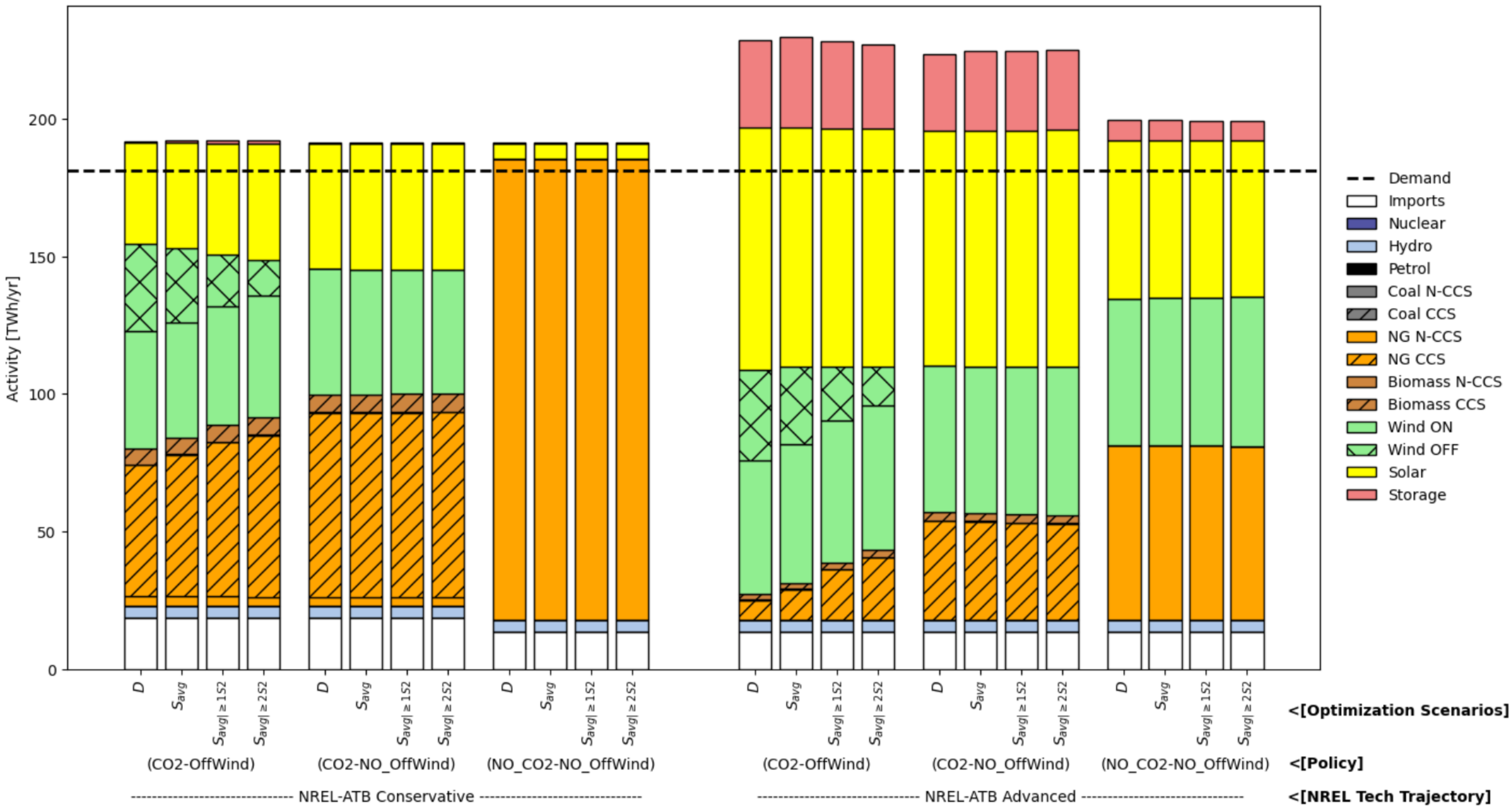
Scenario Tree Construction

Scenario Tree

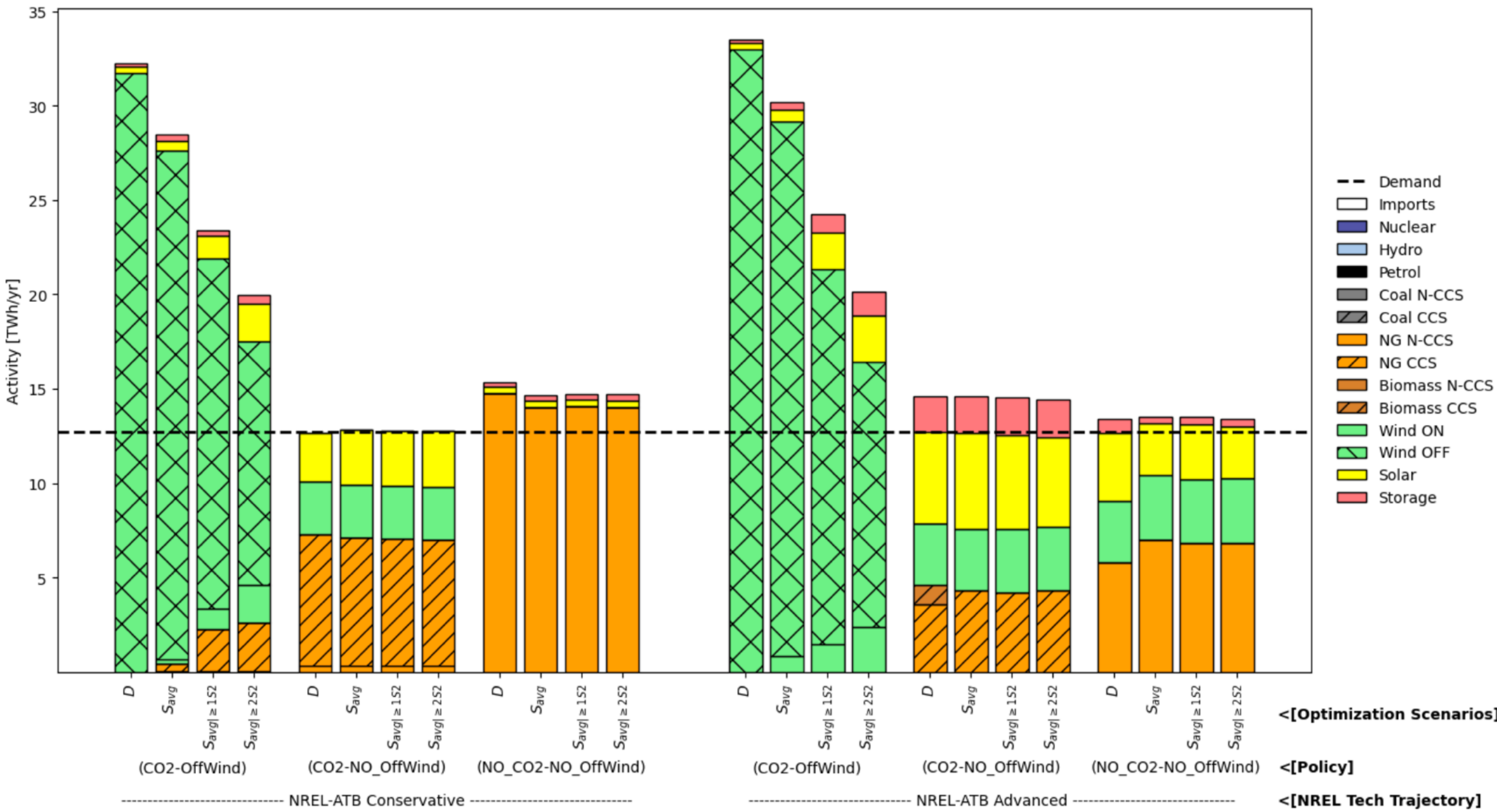


- We have to inform the optimization model of all $3^5 = 243$ Possible Realizations. Grows fast $3^6 = 729$, $3^7 = 2187$ (Curse of Dimensionality)
- At a given time stage and scenario, infrastructure is damaged and new generation needs to be built to meet demand
- We do not evaluate reconstruction time

Activity Results in 2050 - NC



Activity Results in 2050 – NC Region 1



de Faria, V.A.D., Predictive and Prescriptive Analytics Applied to Energy Systems, PhD Dissertation, North Carolina State University, July of 2024

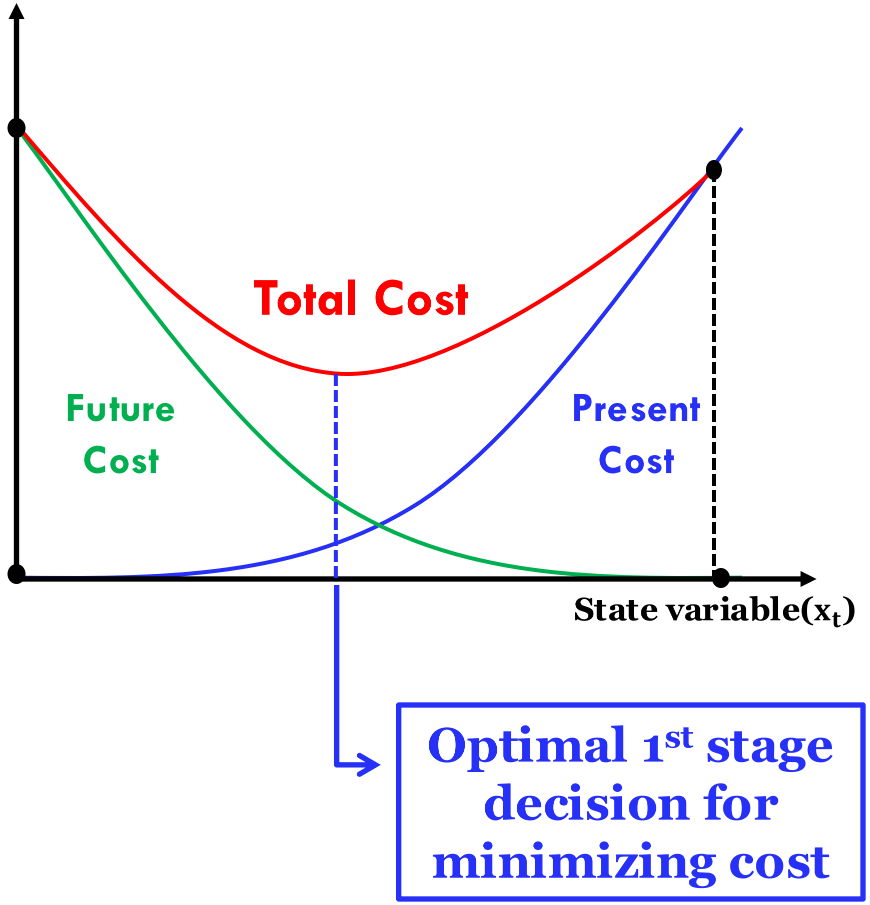
Takeaways – NC Analysis

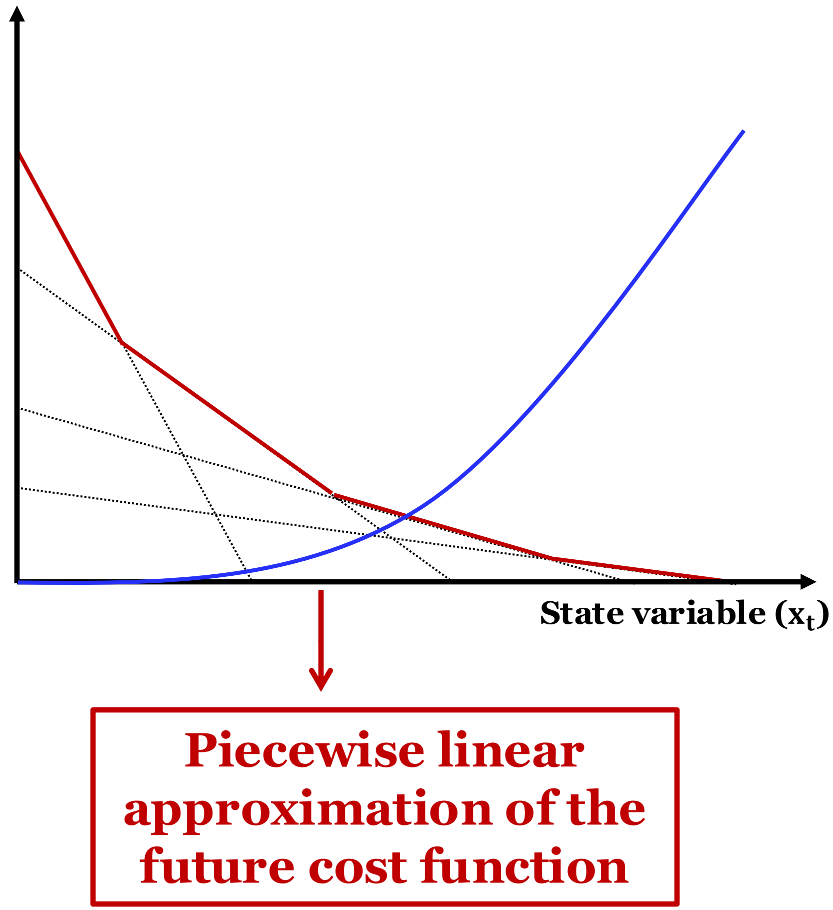
- At present, most of the NC system is composed of fossil fuel technologies, which are resilient against hurricanes
- Our ability up to this point with TEMOA is to consider a limited amount of stages and scenarios due to the computational intractability as the stochastic optimization model gets larger
- The investigation of decomposition strategies in accelerating the simulation of the stochastic optimization model for capacity planning is something of interest

Sampling-based Decomposition

Let's Recall

- **Minimize total costs (present + future)**





Stage-t Benders' Master Problem

- Suppose we are at stage t under ω and we have:

$$\begin{aligned} \min_{x_t,\theta_t} \quad & c_t x_t + \theta_t \\ \text{s.t.} \quad & A_t x_t = B_t x_{t-1} + b_t \quad : \pi_t \\ & -\vec{G}_t x_t + e\,\theta_t \geq \vec{g}_t \quad : \alpha_t \\ & x_t \geq 0 \end{aligned}$$

Benders' cuts

$$b_t = R_{t-1} b_{t-1} + \eta_t$$

$\mathrm{vec}(\eta_t, c_t, B_t, A_t), t = 2, \ldots, T$ **are** $\perp\!\!\!\perp$

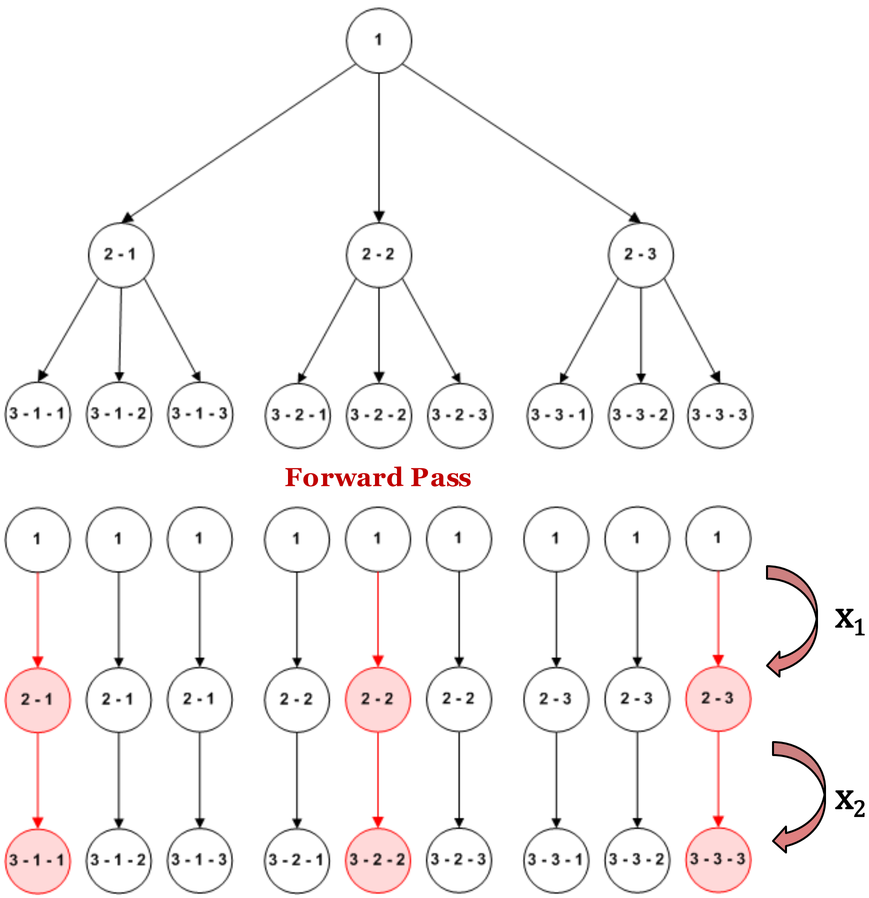
cut-intercept vector

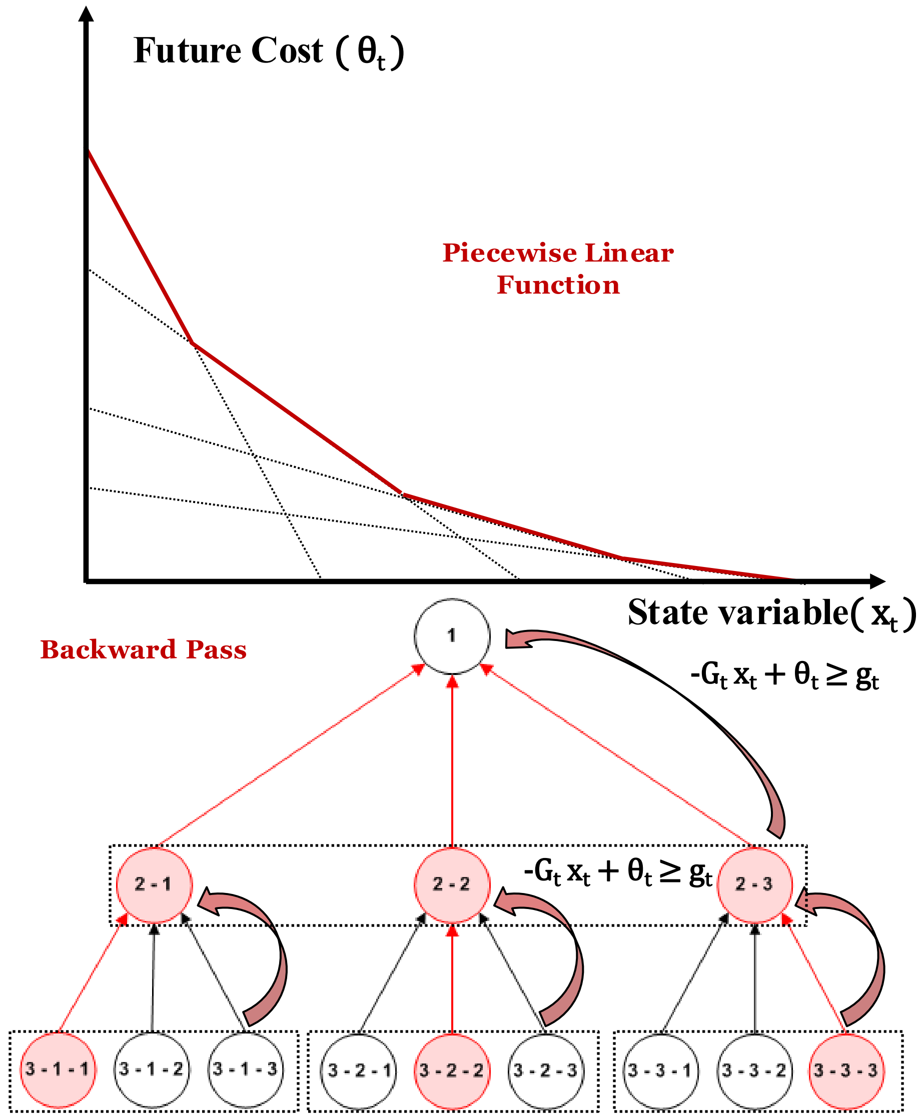
$$G_t = \sum_{\omega_{t+1} \in \Delta(\omega_t)} p^{\omega_{t+1}|\omega_t} \pi_{t+1}^{\omega_{t+1}} B_{t+1}$$

cut-gradient matrix

$$g_t^{\omega_t} = \sum_{\omega_{t+1} \in \Delta(\omega_t)} p^{\omega_{t+1}|\omega_t} \pi_{t+1}^{\omega_{t+1}} b_{t+1}^{\omega_{t+1}} + \sum_{\omega_{t+1} \in \Delta(\omega_t)} p^{\omega_{t+1}|\omega_t} \alpha_{t+1}^{\omega_{t+1}} \vec{g}_{t+1}^{\,\omega_{t+1}}$$

Sampling-based Decomposition Algorithm

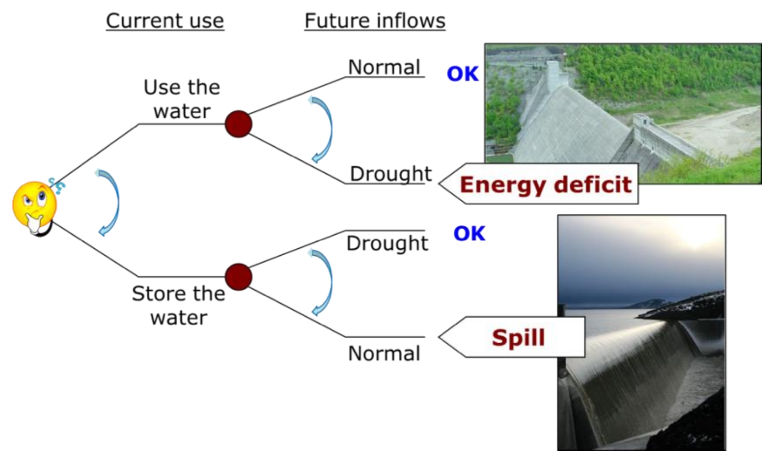




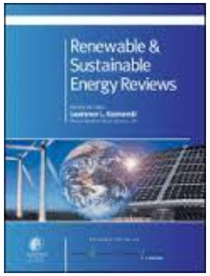
de Queiroz, A.R., Morton, D.P., (2013) Sharing Cuts under Aggregated Forecasts when Decomposing Multi-stage Stochastic Programs, Operations Research Lett, 41(3): 311-316

Stochastic Hydro-thermal Scheduling

- Find the sequence of **hydro releases** and **thermal plant dispatches** for a planning horizon to match system demand
  - Resource management
  - Input variable forecasting
  - Operational aspects



- Basic economic criterion
  - **Minimize operational costs** (present + expected future)
- Usually modeled as Multi-stage Stochastic Linear Program (SLP-*t*)



de Queiroz, A.R., (2016) Stochastic Hydro-thermal Scheduling Optimization: An Overview, Renewable and Sustainable Energy Reviews, 62: 382-395

HTSP Model Formulation for Stage-t

$$h_t(x^{t-1}, b_t^{\omega}) = \min \overbrace{\sum_{\ell \in L} c_\ell^t GT_\ell^t + \sum_{k \in K} u_k^t GD_k^t}^{\textbf{Present Cost}} + \overbrace{\frac{1}{(1+\beta)} \mathbb{E}_{\,b_{t+1}|b_1,\dots,b_t} h_{t+1}(x^t, b_{t+1})}^{\textbf{Expected Future Cost}}$$

Water Balance

$$\text{s.t.} \quad x_i^t + GH_i^t + S_i^t - \sum_{j \in M_i} \left(GH_j^t + S_j^t\right) = x_i^{t-1} + b_{t+1}^{\omega} \quad \forall i \in I$$

Demand Satisfaction

$$\sum_{i \in I_r} \rho_i GH_i^t + \sum_{\ell \in L} GT_\ell^t + \sum_{k \in K} GD_k^t - \sum_{\substack{r' \in R \\ r' \neq r}} F_{r\,r'}^t + \sum_{\substack{r' \in R \\ r' \neq r}} F_{r\,r'}^t = D_{tr} \quad \forall r \in R$$

$$\underline{x}_i^t \leq x_i^t \leq \bar{x}_i^t \qquad \forall i \in I$$

$$0 \leq GH_i^t \leq \overline{GH}_i^t \qquad \forall i \in I$$

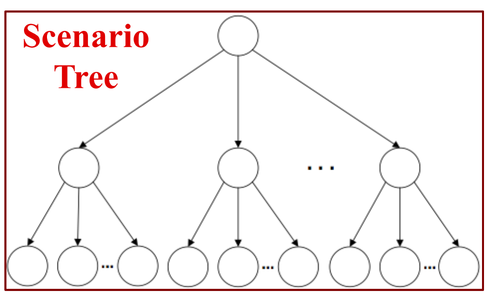
$$0 \leq S_i^t \qquad \forall i \in I$$

$$\underline{GT}_\ell^t \leq GT_\ell^t \leq \overline{GT}_\ell^t \qquad \forall \ell \in L$$

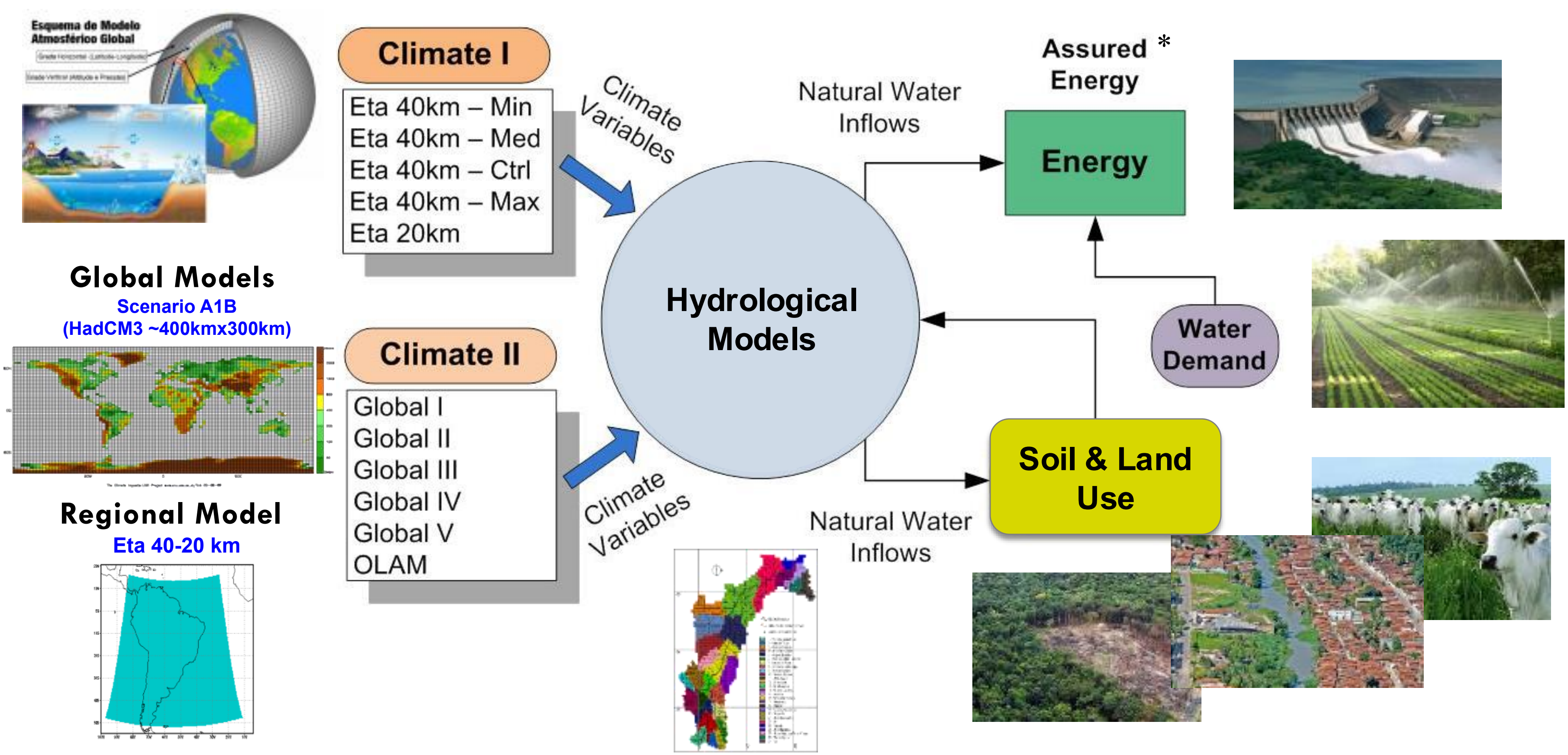
$$0 \leq GD_k^t \qquad \forall k \in K$$

$$0 \leq F_{r\,r'}^t \leq \overline{F}_{r\,r'}^t \qquad \forall (r, r') \in R$$

Simple Bounds



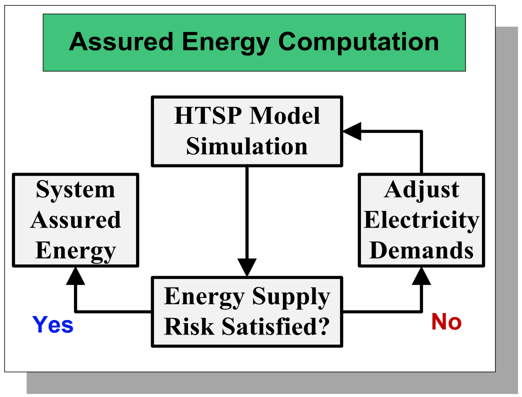
Water-Energy Nexus Under Changing Climate

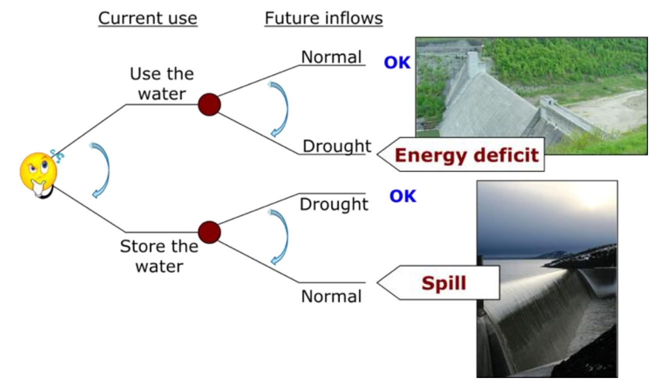


The system Assured Energy is a hypothetical amount of energy that could be generated by a power generation system for a pre-determined level of supply risk

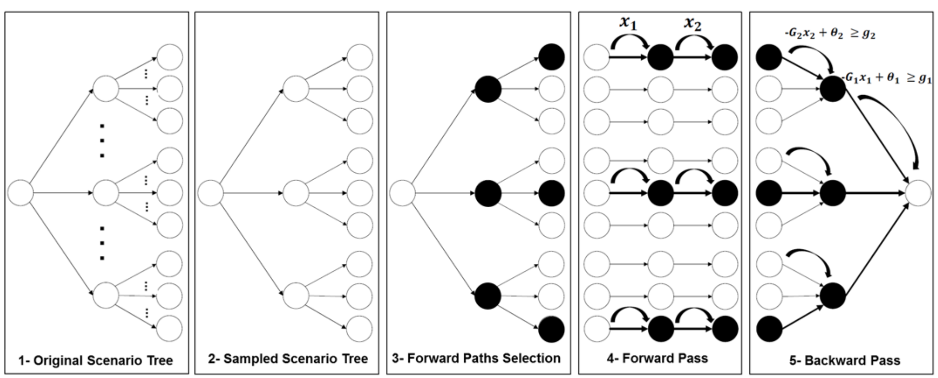
HTSP for Assured Energy Assessment under Climate Change

- To define the system Assured Energy we need a **simulation-optimization framework** based on the hydro-thermal scheduling problem (HTSP)

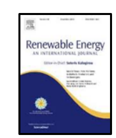




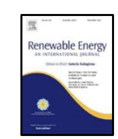
- A **stochastic programming** model for the **HTSP** is used to identify the system's potential (assured energy)



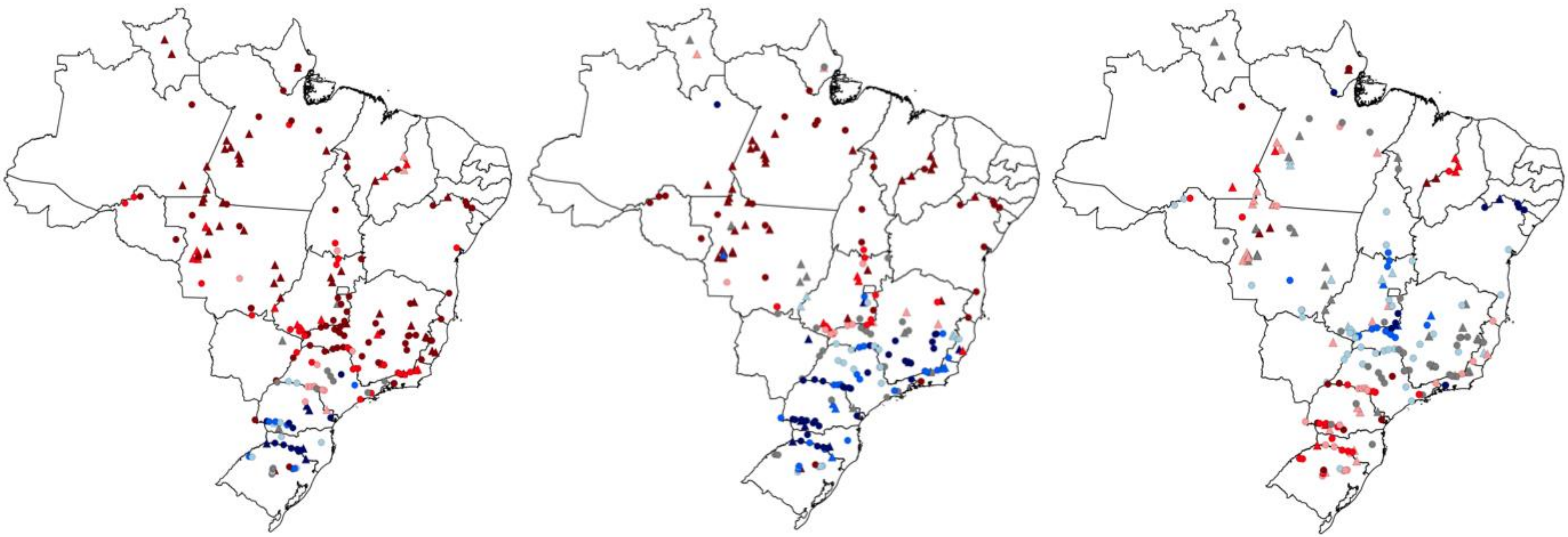
de Queiroz, A.R., Lima, L.M.M., Lima, J.W.M., Silva, B.C., Scianni, L.A., (2016) Climate Change Impacts in the Energy Supply of the Brazilian Hydro-dominant Power System, Renewable Energy, 99: 379-389



de Queiroz, A. R., Faria, V. A., Lima, L. M., & Lima, J. W. (2019). Hydropower revenues under the threat of climate change in Brazil. Renewable energy, 133, 873-882



Do These new Projects Make Sense?



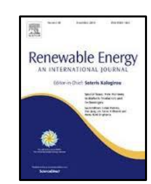
P → F1   P → F2   P → F3

Hydro plant Exist - New	Percentage Changes in AE	Number of hydro plants		
		P - F1	F1-F2	F2-F3
● - ▲	<-25%	63-43	28-35	9-5
● - ▲	-25% to -15%	30-17	11-6	19-9
● - ▲	-15% to -5%	10-3	7-5	21-21
● - ▲	-5% to 5%	11-3	22-8	34-25
● - ▲	5% to 15%	8-1	18-4	36-8
● - ▲	15% to 25%	7-0	20-6	13-3
● - ▲	>25%	12-6	35-9	9-2

The answer we got at the individual level using is that most of these new projects do not make sense!

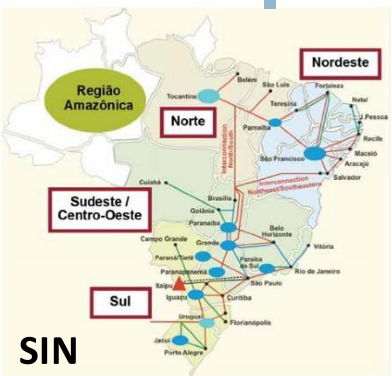
Considering a marginal expansion cost of 54$/MWh ***the annual losses are $5(12) billions EGP (FGP) from P to F1***

de Queiroz, A.R., de Faria, V.A.D., Lima, L.M.M., Lima, J.W.M., (2019) Hydro Power Revenues Under the Threat of Climate Change in Brazil, Renewable Energy, 133: 873-882

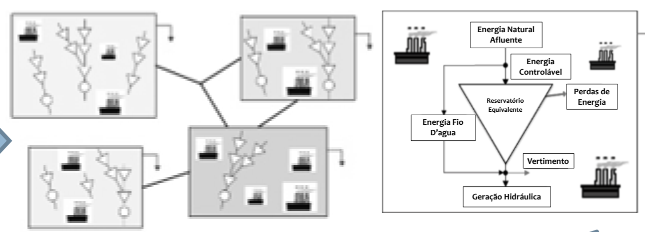


TEMOA & Storage in the Brazilian System

- Hydro-dominant Power System (~70% of production)
- Individual Hydro and Thermal Plants Representation
- Interconnection between Regions
- De-rating factor, efficiency, capacity factor, and transmission losses

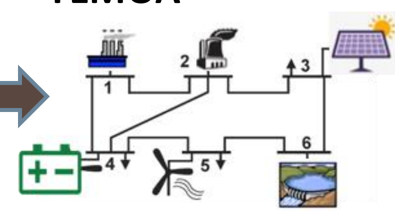


NEWAVE



Monthly Hydro Generation by Aggregated Reservoir

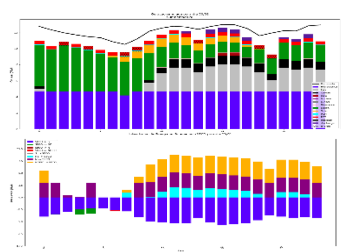
TEMOA

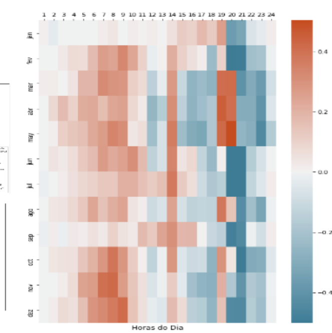


Constraints

Hourly information by region

- Wind
- Solar
- Demand





Outputs:

- Generation dispatch per plant
- Storage Charge/Discharge
- Energy Exchanges between Regions
- Total Operational Costs

Final Comments

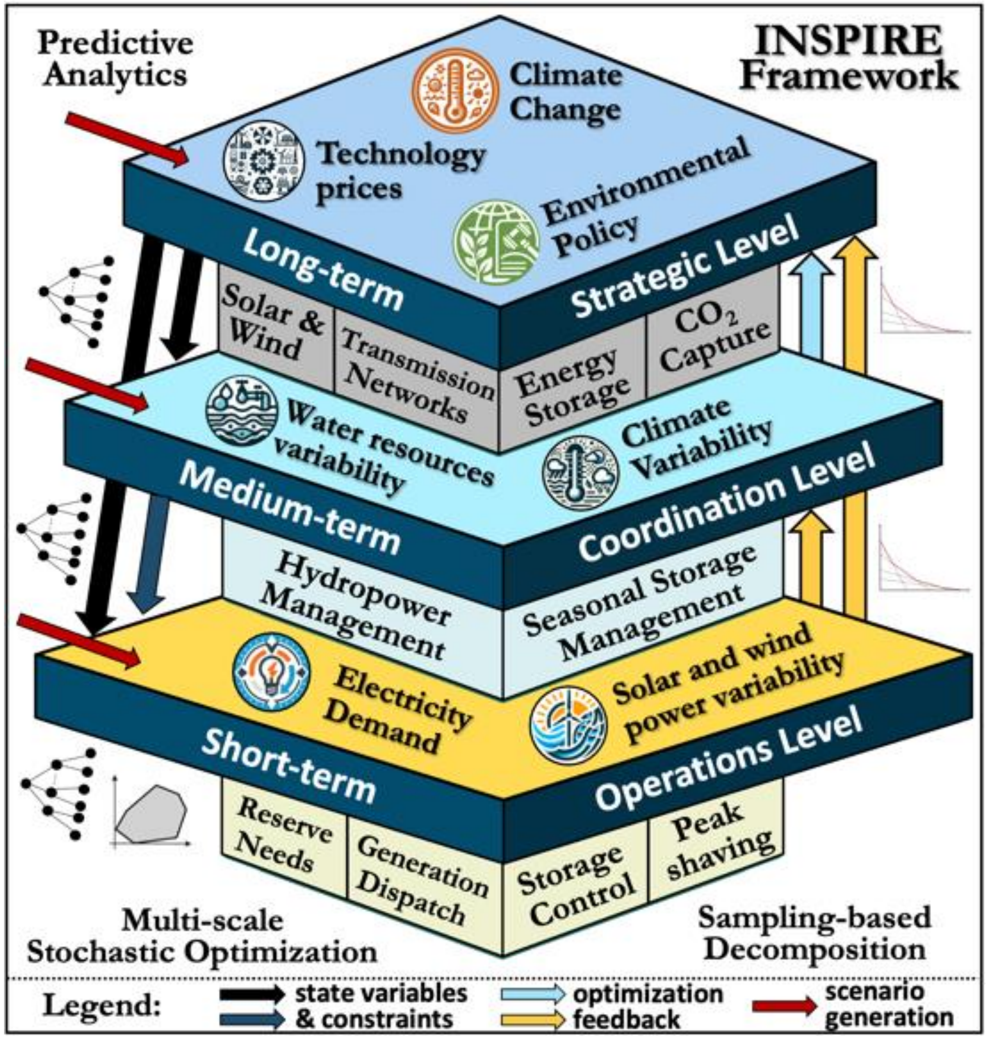
Final Comments

- We presented an introduction to stochastic programming with applications to energy and power
- We discussed about **multi-stage stochastic programming and sampling-based decomposition** used to solved such models
- We discussed how TEMOA can represent stochastic optimization problems
- A framework based on **stochastic hydrothermal scheduling** to provide useful information and support decision-making in problems related to investments in hydropower was shown
- Future directions should explore **multi-scale aspects of the problems** due to different granularity of the uncertainty in renewables

Future Work

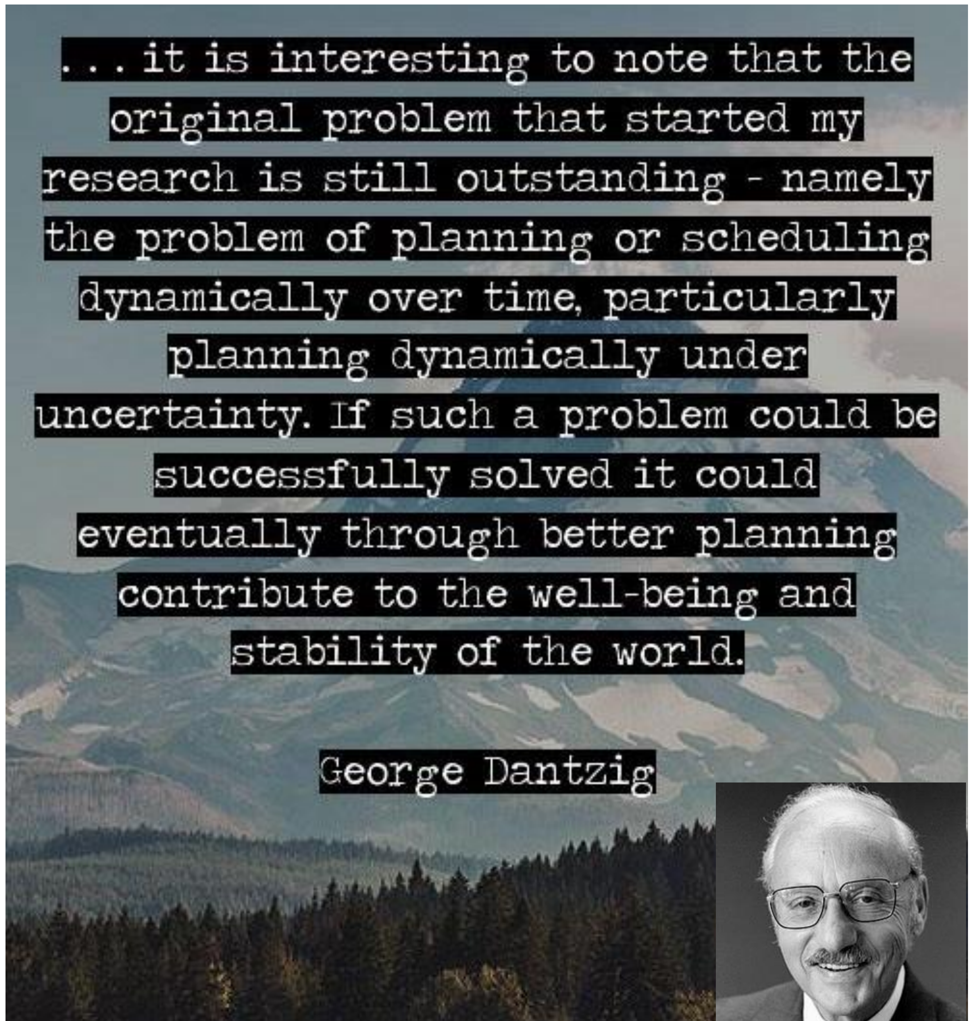
INtegrated Stochastic optimization and Predictive analytics In REsilient multi-scale systems (INSPIRE)

- Framework to jointly consider **strategic & operational uncertainties**
- **Multi-scale optimization**
- **Predictive analytics** to generate scenario trees
- Develop and tailor a parallel sampling-based decomposition algorithm to solve multi-scale problems



To Wrap up

In 1991…



… after 34 years it is still outstanding…

MOLTE GRAZIE !

ardequei@ncsu.edu

@ar_queiroz

ar-queiroz

https://ardequeiroz.github.io

How to Modify TEMOA to Handle Stochastic Optimization

TEMOA as a Stochastic Optimization Model: The Basics

You Need to Make Modifications Tailored to Your Case

- Need to inform the model what is the stochastic parameter we are interest (in our case CapacityReduction), and how it affects the model constraints

Add new variable on temoa_model.py

```
#Capacity Reduction
M.CapReduction = Param( M.time_optimize, M.tech_all, M.vintage_all, default = 1.0)
```

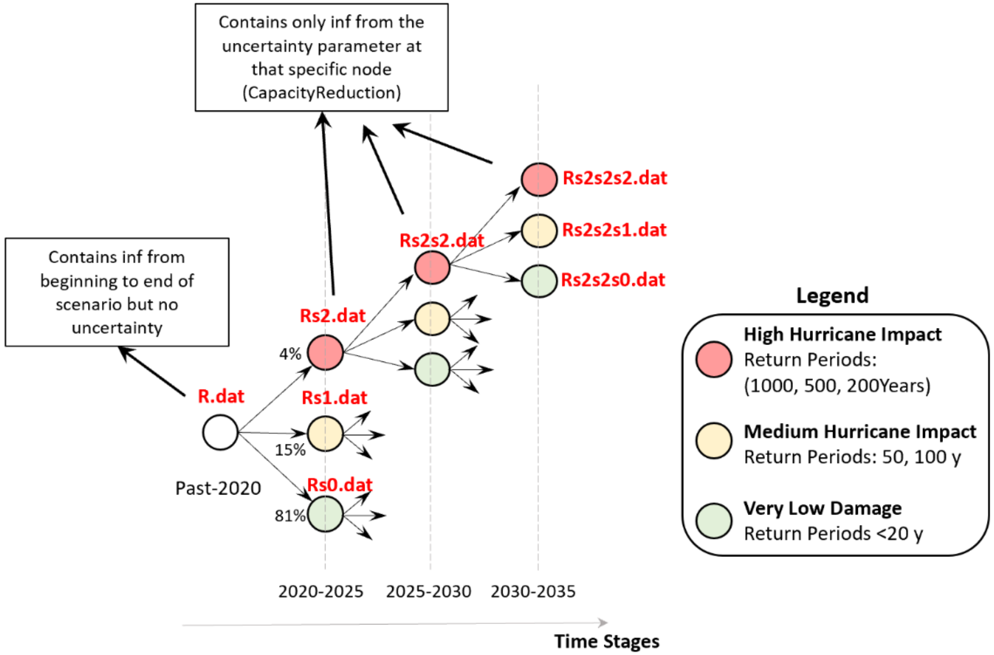
Modifications on temoa_rules.py

```
def availableActivity_Constraint(M, p, t, v):
    # This is max produceable in a given year by a perticular technology
    max_produceable = sum((
                              (value(M.CapacityFactorProcess[S_s, S_d, t, S_v])
                               * value(M.CapacityToActivity[t])
                               * value(M.SegFrac[S_s, S_d]))
                              * value(M.ProcessLifeFrac[p, t, S_v])
                              * M.V_Capacity[t, S_v]
                              * value(M.CapReduction[p, t, S_v]))  #Hurricane  Model
                        for S_s in M.time_season
                        for S_d in M.time_of_day
                        for S_v in M.ProcessVintages(p, t))

    # This is what is produced in a year by a particular technology
    activity_pt = sum(M.V_Activity[p, S_s, S_d, t, S_v]
                    for S_s
                    for S_d
                    for S_v

        expr = (activity_pt <
    return expr
```

```
def Capacity_Constraint(M, p, s, d, t, v):
    r"""
    if t in M.tech_storage:
    # The expressions below are defined in-line to minimize the amount of
    # expression cloning taking place with Pyomo.
    if t in M.tech_curtailment:
    else:
        return value(M.CapacityFactorProcess[s, d, t, v]) \
        * value(M.CapacityToActivity[t]) \
        * value(M.SegFrac[s, d]) \
        * value(M.ProcessLifeFrac[p, t, v]) \
        * M.V_Capacity[t, v] \
        * value(M.CapReduction[p, t, v])  >= M.V_Activity[p, s, d, t, v] #Hurricane  Model
```

TEMOA as a Stochastic Optimization Model: The Root Node

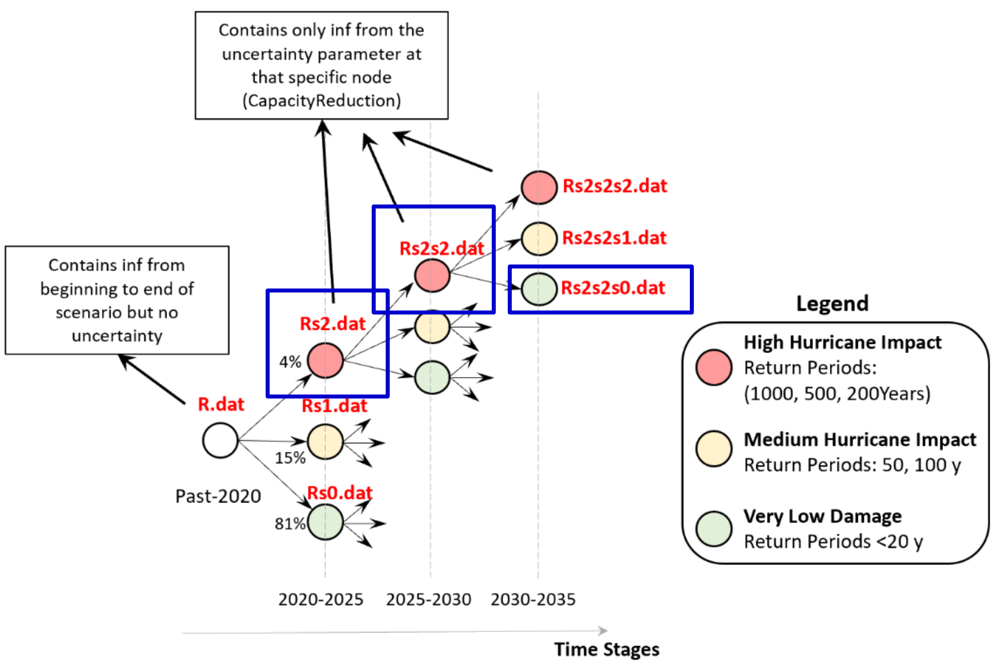
- As each problem may have a different set of stochastic parameters we need to create the necessary code to inform the scenarios for Pyomo
- Define the parameters of the root node (problem without uncertainty- R.dat > ReferenceModel.dat). This is the traditional deterministic TEMOA files, without the stochastic parameter

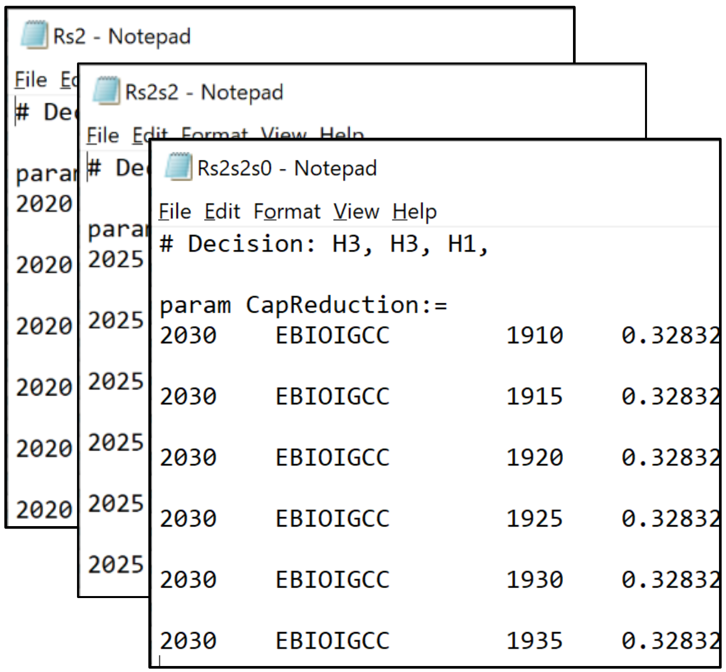


TEMOA as a Stochastic Optimization Model: The Definition of Scenarios

- As each problem may have a different set of stochastic parameters we need to create the necessary code to inform the scenarios for Pyomo

- Define the parameters of each individual node (Rs0.dat, Rs0s1.dat ,…)

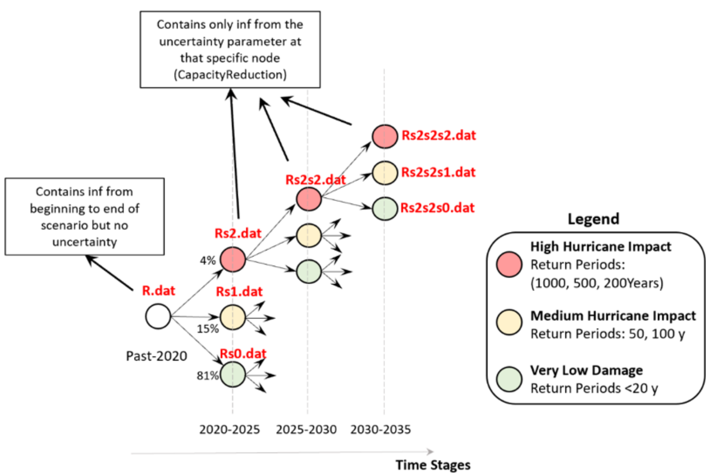




TEMOA as a Stochastic Optimization Model: The Scenario Structure

- As each problem may have a different set of stochastic parameters we need to create the necessary code to inform the scenarios for Pyomo

- Define how each node connect with one another (ScenarioStructure.dat)



ScenarioStructure - Notepad

File Edit Format View Help

```
set  Scenarios  :=
  S0s0s0
  S0s0s1
  S0s0s2
  S0s1s0
  S0s1s1
  S0s1s2
  S0s2s0
  S0s2s1
  S0s2s2
  S1s0s0
  S1s0s1
  S1s0s2
  S1s1s0
  S1s1s1
  S1s1s2
  S1s2s0
  S1s2s1
  S1s2s2
  S2s0s0
  S2s0s1
  S2s0s2
  S2s1s0
  S2s1s1
```

ScenarioStructu

File Edit Format V

```
set  Nodes  :
  R
  Rs0
  Rs0s0
  Rs0s0s0
  Rs0s0s1
  Rs0s0s2
  Rs0s1
  Rs0s1s0
  Rs0s1s1
  Rs0s1s2
  Rs0s2
  Rs0s2s0
  Rs0s2s1
  Rs0s2s2
  Rs1
  Rs1s0
  Rs1s0s0
  Rs1s0s1
  Rs1s0s2
  Rs1s1
  Rs1s1s0
  Rs1s1s1
```

ScenarioStructure - Notepad

File Edit Format View Help

```
set  Children[Rs1s2]  :=
  Rs1s2s0
  Rs1s2s1
  Rs1s2s2

set  Ch
  Rs2s
  Rs2s
  Rs2s

set  C
  Rs2s0s0
  Rs2s0s1
  Rs2s0s2
      ;

set  Children[Rs2s1]
  Rs2s1s0
  Rs2s1s1
  Rs2s1s2
      ;
```

ScenarioStructure - Notepad

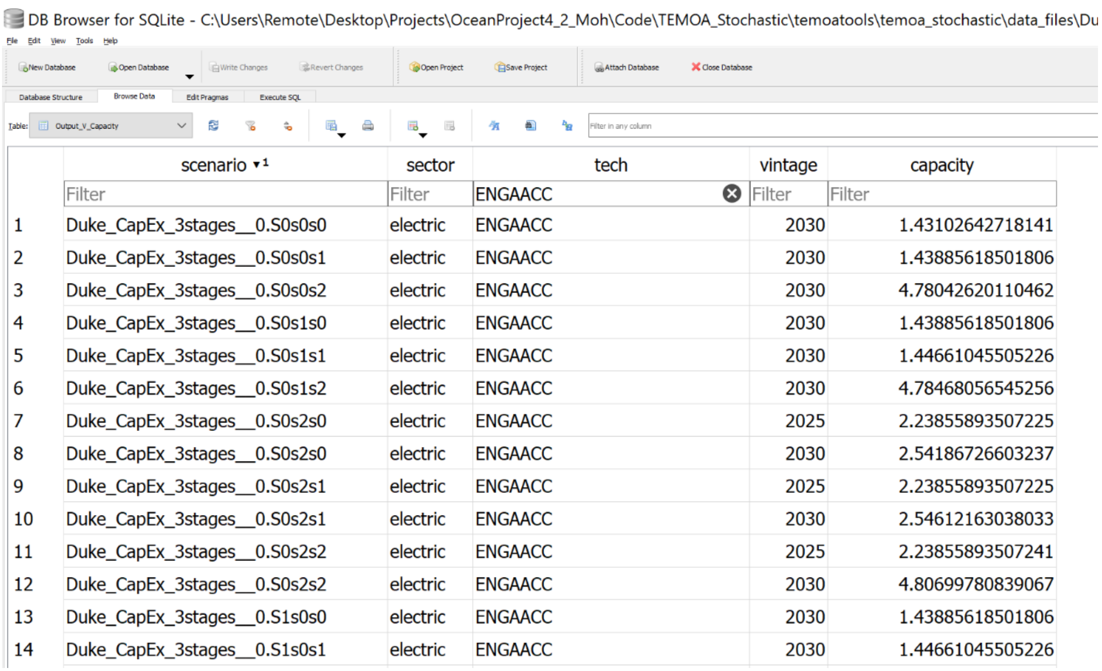
File Edit Format View Help

```
set  StageVariables[s2020]  :=
  V_FlowIn[2020,s001,tod01,COABH_IGCC_N,E_BLND_BITHML_COALIGCC_N,2017,COALIGCC_N_B]
  V_FlowIn[2020,s001,tod01,COABH_IGCC_N,E_BLND_BITHML_COALIGCC_N,2020,COALIGCC_N_B]
  V_FlowIn[2020,s001,tod01,COABH_N,E_BLND_BITHML_COALSTM_N,2017,COALSTM_N_B]
  V_FlowIn[2020,s001,tod01,COABH_N,E_BLND_BITHML_COALSTM_N,2020,COALSTM_N_B]
  V_FlowIn[2020,s001,tod01,COABH_R,E_FGD_COABH_N,2017,COAB_R_CC]
  V_FlowIn[2020,s                                          R_CC]
  V_FlowIn[2020,s                                          R_CC]
```

```
param  ConditionalProbability  :=
  R  1
  Rs0  0.81
  Rs0s0  0.81
  Rs0s0s0  0.81
  Rs0s0s1  0.15
  Rs0s0s2  0.0.04
  Rs0s1  0.15
  Rs0s1s0  0.81
  Rs0s1s1  0.15
  Rs0s1s2  0.04
```

TEMOA as a Stochastic Optimization Model: Creating the Files & Optimizing

- Create scenarios → Create scenario structure → Run (temoa_stochastic.py) → Results



DB Browser for SQLite - C:\Users\Remote\Desktop\Projects\OceanProject4_2_Moh\Code\TEMOA_Stochastic\temoatools\temoa_stochastic\data_files\Du

	scenario	sector	tech	vintage	capacity
	Filter	Filter	ENGAACC	Filter	Filter
1	Duke_CapEx_3stages__0.S0s0s0	electric	ENGAACC	2030	1.43102642718141
2	Duke_CapEx_3stages__0.S0s0s1	electric	ENGAACC	2030	1.43885618501806
3	Duke_CapEx_3stages__0.S0s0s2	electric	ENGAACC	2030	4.78042620110462
4	Duke_CapEx_3stages__0.S0s1s0	electric	ENGAACC	2030	1.43885618501806
5	Duke_CapEx_3stages__0.S0s1s1	electric	ENGAACC	2030	1.44661045505226
6	Duke_CapEx_3stages__0.S0s1s2	electric	ENGAACC	2030	4.78468056545256
7	Duke_CapEx_3stages__0.S0s2s0	electric	ENGAACC	2025	2.23855893507225
8	Duke_CapEx_3stages__0.S0s2s0	electric	ENGAACC	2030	2.54186726603237
9	Duke_CapEx_3stages__0.S0s2s1	electric	ENGAACC	2025	2.23855893507225
10	Duke_CapEx_3stages__0.S0s2s1	electric	ENGAACC	2030	2.54612163038033
11	Duke_CapEx_3stages__0.S0s2s2	electric	ENGAACC	2025	2.23855893507241
12	Duke_CapEx_3stages__0.S0s2s2	electric	ENGAACC	2030	4.80699780839067
13	Duke_CapEx_3stages__0.S1s0s0	electric	ENGAACC	2030	1.43885618501806
14	Duke_CapEx_3stages__0.S1s0s1	electric	ENGAACC	2030	1.44661045505226

- In the stochastic model the solution is dependent on which hurricane scenario is realized